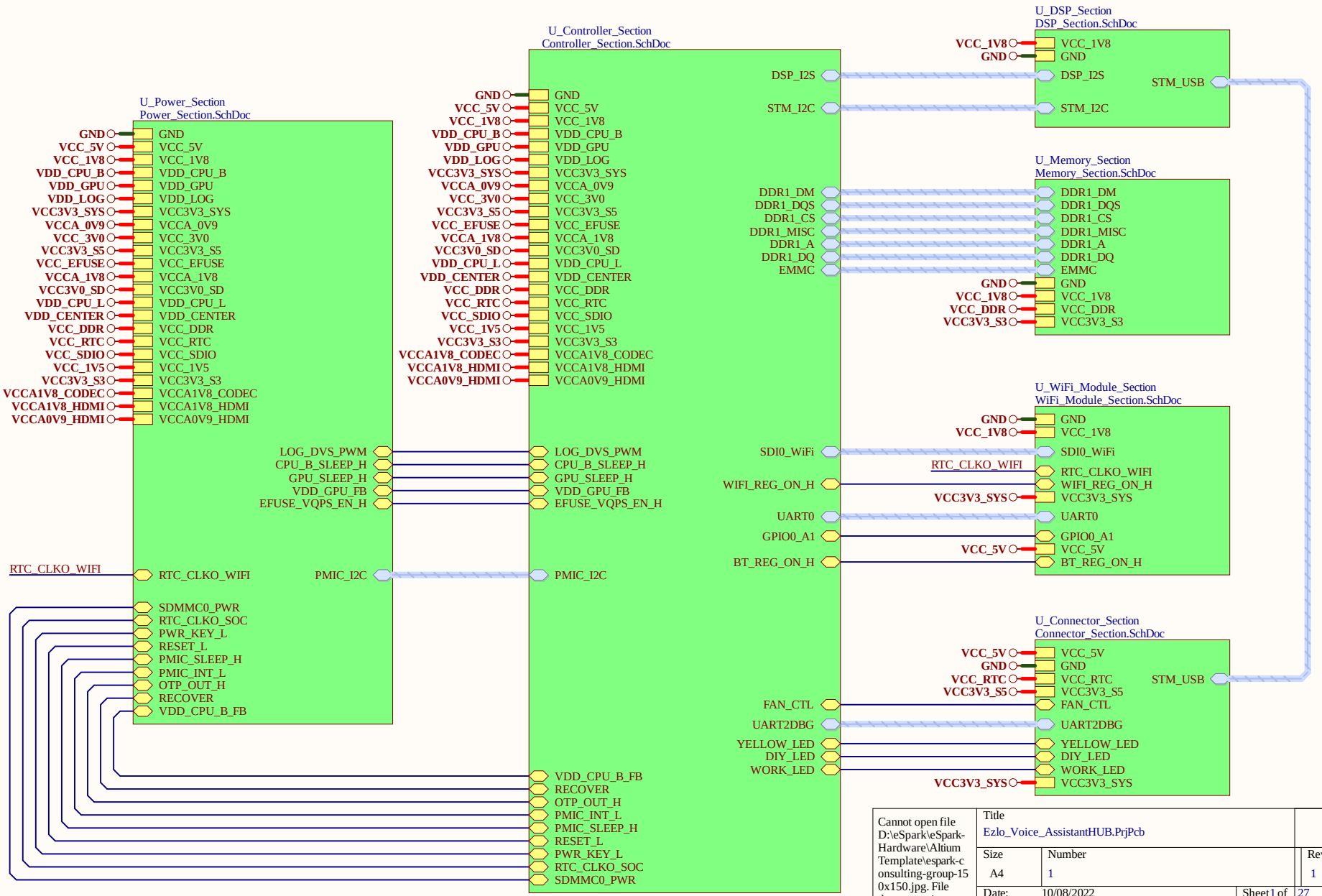
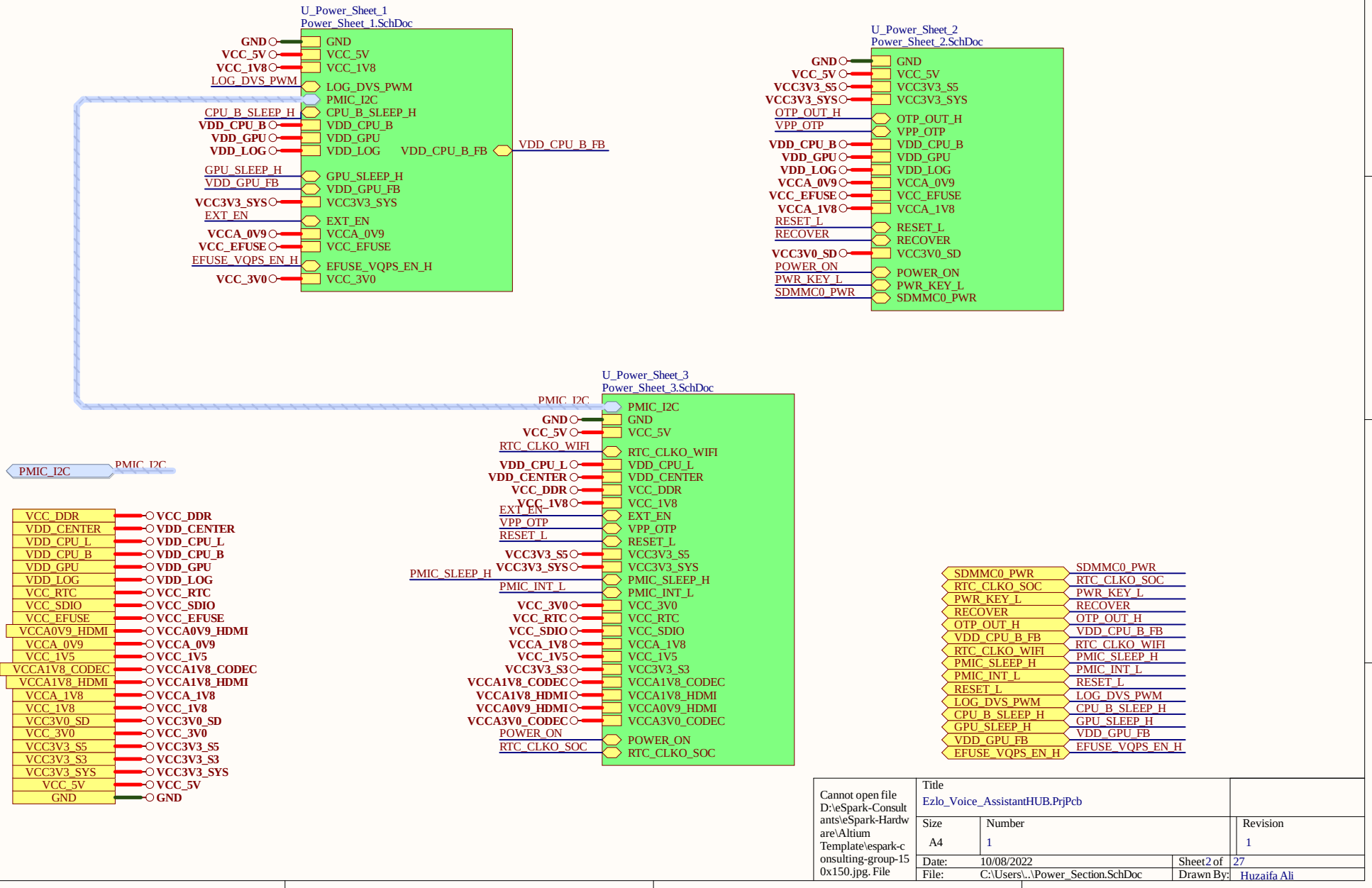
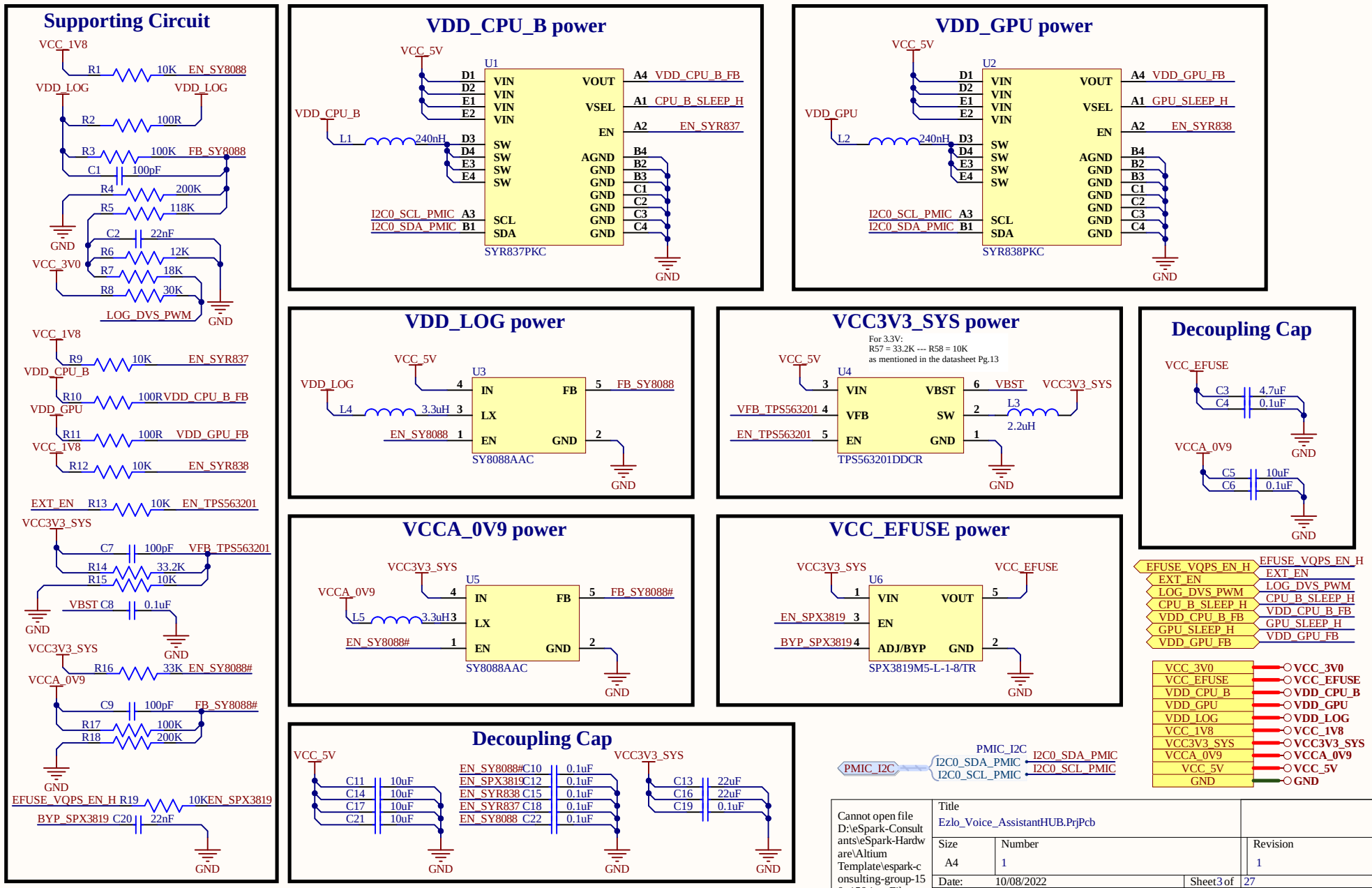
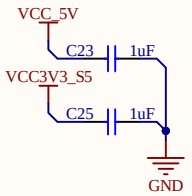

U_Power_Section
Power_Section.SchDoc
GND
VCC_5V
VCC_1V8
VDD_CPU_B
VDD_GPU
VDD_LOG
VCC3V3_SYS
VCCA_0V9
VCC_3V0
VCC3V3_S5
VCC_EFUSE
VCCA_1V8
VCC3V0_SD
VDD_CPU_L
VDD_CENTER
VCC_DDR
VCC_RTC
VCC_SDIO
VCC_1V5
VCC3V3_S3
VCCA1V8_CODEC
VCCA1V8_HDMI
VCCA0V9_HDMI
LOG_DVS_PWM
CPU_B_SLEEP_H
GPU_SLEEP_H
VDD_GPU_FB
EFUSE_VQPS_EN_H
RTC_CLKO_WIFI
PMIC_I2C
SDMMC0_PWR
RTC_CLKO_SOC
PWR_KEY_L
RESET_L
PMIC_SLEEP_H
PMIC_INT_L
OTP_OUT_H
RECOVER
VDD_CPU_B_FB
U_Controller_Section
Controller_Section.SchDoc
DSP_I2S
STM_I2C
U_DSP_Section
DSP_Section.SchDoc
STM_USB
U_Memory_Section
Memory_Section.SchDoc
DDR1_DM
DDR1_DQS
DDR1_CS
DDR1_MISC
DDR1_A
DDR1_DQ
EMMC
U_WiFi_Module_Section
WiFi_Module_Section.SchDoc
SDI0_WiFi
WIFI_REG_ON_H
UART0
GPIO0_A1
BT_REG_ON_H
U_Connector_Section
Connector_Section.SchDoc
FAN_CTL
UART2DBG
YELLOW_LED
DIY_LED
WORK_LED
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size
A4
Number
1
Revision
1
Date:
10/08/2022
Sheet 1 of 27
File:
C:\Users\..\Top_Sheet.SchDoc
Drawn By:
Huzaifa Ali



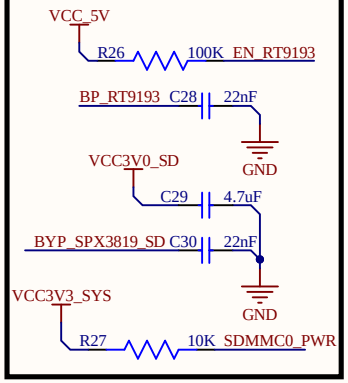
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	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 2 of 27	
	File: C:\Users\..\Power_Section.SchDoc	Drawn By: Huzaifa Ali	


Supporting Circuit
VCC_1V8
R1 10K EN_SY8088
VDD_LOG
VDD_LOG
R2 100R
R3 100K FB_SY8088
C1 100pF
R4 200K
R5 118K
GND
C2 22nF
R6 12K
VCC_3V0
R7 18K
R8 30K
LOG_DVS_PWM
GND
VCC_1V8
R9 10K EN_SY837
VDD_CPU_B
R10 100R VDD_CPU_B_FB
VDD_GPU
R11 100R VDD_GPU_FB
VCC_1V8
R12 10K EN_SY838
EXT_EN R13 10K EN_TPS563201
VCC3V3_SYS
C7 100pF VFB_TPS563201
R14 33.2K
R15 10K
VBST C8 0.1uF
GND
GND
VCC3V3_SYS
R16 33K EN_SY8088#
VCCA_0V9
C9 100pF FB_SY8088#
R17 100K
R18 200K
GND
EFUSE_VQPS_EN_H R19 10K EN_SPX3819
BYP_SPX3819 C20 22nF
GND
VDD_CPU_B power
VCC_5V
U1
D1 VIN
D2 VIN
E1 VIN
E2 VIN
VOUT A4 VDD_CPU_B_FB
VSEL A1 CPU_B_SLEEP_H
EN A2 EN_SYR837
VDD_CPU_B
L1 240nH
D3 SW
D4 SW
E3 SW
E4 SW
AGND B4
GND B2
GND B3
GND C1
GND C2
GND C3
GND C4
I2C0_SCL_PMIC A3
I2C0_SDA_PMIC B1
SCL
SDA
SYR837PKC
GND
VDD_GPU power
VCC_5V
U2
VOUT A4 VDD_GPU_FB
VSEL A1 GPU_SLEEP_H
EN A2 EN_SYR838
VDD_GPU
L2 240nH
I2C0_SCL_PMIC A3
I2C0_SDA_PMIC B1
SYR838PKC
GND
VDD_LOG power
VCC_5V
U3
VDD_LOG
4 IN
FB 5 FB_SY8088
L4 3.3uH 3
LX
EN_SY8088 1
EN
GND 2
SY8088AAC
GND
VCC3V3_SYS power
For 3.3V:
R57 = 33.2K --- R58 = 10K
as mentioned in the datasheet Pg.13
VCC_5V
U4
3 VIN
VBST 6 VBST
VCC3V3_SYS
VFB_TPS563201 4
VFB
SW 2
L3
2.2uH
EN_TPS563201 5
EN
GND 1
TPS563201DDCR
GND
Decoupling Cap
VCC_EFUSE
C3 4.7uF
C4 0.1uF
GND
VCCA_0V9
C5 10uF
C6 0.1uF
GND
VCCA_0V9 power
VCC3V3_SYS
U5
VCCA_0V9
4 IN
FB 5 FB_SY8088#
L5 3.3uH 3
LX
EN_SY8088# 1
EN
GND 2
SY8088AAC
GND
VCC_EFUSE power
VCC3V3_SYS
U6
1 VIN
VOUT 5 VCC_EFUSE
EN_SPX3819 3
EN
BYP_SPX3819 4
ADJ/BYP
GND 2
SPX3819M5-L-1-8/TR
GND
EFUSE_VQPS_EN_H
EXT_EN
LOG_DVS_PWM
CPU_B_SLEEP_H
VDD_CPU_B_FB
GPU_SLEEP_H
VDD_GPU_FB
Decoupling Cap
VCC_5V
C11 10uF
C14 10uF
C17 10uF
C21 10uF
GND
EN_SY8088# C10 0.1uF
EN_SPX3819 C12 0.1uF
EN_SYR838 C15 0.1uF
EN_SYR837 C18 0.1uF
EN_SY8088 C22 0.1uF
GND
VCC3V3_SYS
C13 22uF
C16 22uF
C19 0.1uF
GND
PMIC_I2C
PMIC_I2C
I2C0_SDA_PMIC
I2C0_SCL_PMIC
I2C0_SDA_PMIC
I2C0_SCL_PMIC
VCC_3V0
VCC_EFUSE
VDD_CPU_B
VDD_GPU
VDD_LOG
VCC_1V8
VCC3V3_SYS
VCCA_0V9
VCC_5V
GND
VCC_3V0
VCC_EFUSE
VDD_CPU_B
VDD_GPU
VDD_LOG
VCC_1V8
VCC3V3_SYS
VCCA_0V9
VCC_5V
GND
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size
A4
Number
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Revision
1
Date: 10/08/2022
Sheet 3 of 27
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Drawn By: Huzaifa Ali

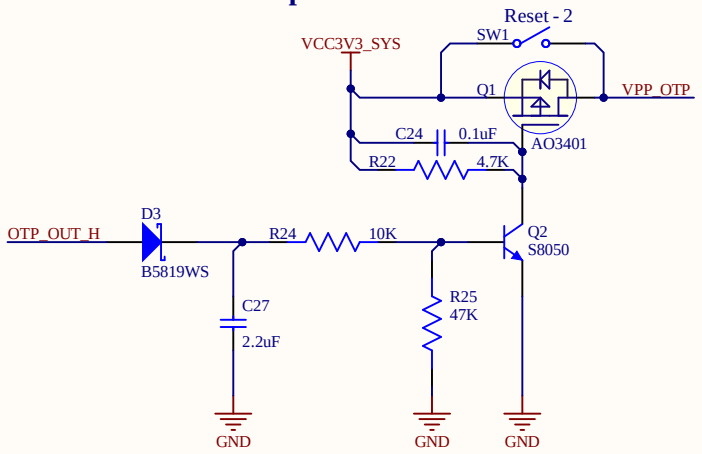
Decoupling Cap



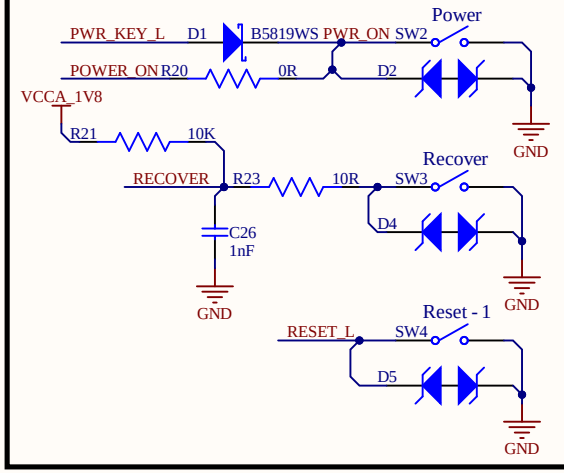
Supporting Circuit



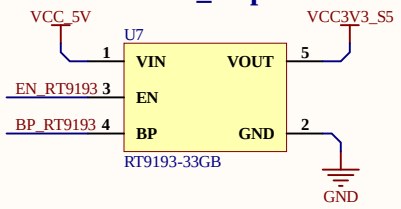
Over-Temperature Protection



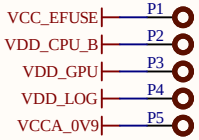
Reset - Power - Recover Keys



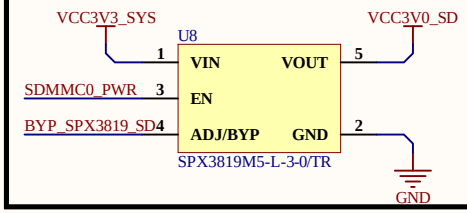
VCC3V3_S5 power

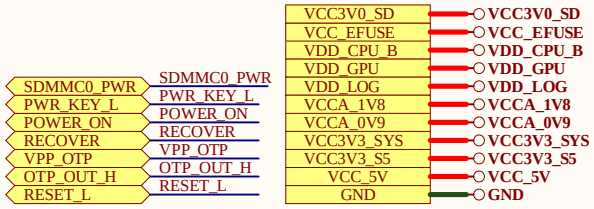


Test Points



VCC3V0_SD power





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Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	10/08/2022	Sheet 4 of	27
File:	C:\Users\..\Power_Sheet_2.SchDoc	Drawn By:	Huzaifa Ali

PMIC_I2C
I2C0_SDA_PMIC
I2C0_SCL_PMIC

VCCA3V0_CODEC
RTC_CLKO_SOC
POWER_ON

EXT_EN
VPP_OTP
RESET_L

RTC_CLKO_WIFI
PMIC_SLEEP_H
PMIC_INT_L

Supporting Circuit

VCC_5V, ESD1, AZ5825-01F, C33 10uF, GND
VCC_VDC, VCC_5V, C34 10uF, D6 B5819WS, GND
R28 150K, R29 270K, VCC_1V5
R32 10K
VCC_DDR, VFB3# C43 47pF, VFB3, R33 51K, R34 200K, VCC_DDR, R37 100R, GND
VPP_OTP, R38 1K, VCC3V3_S5, R39 10K POWER_ON, GND
VCC_1V8, RESET_L R40 10K

Power Management IC RK808

U9A

VCC_5V
VCC3V3_SYS
VCC3V3_S5, VCC_RTC
VDC_RK8, C45 0.1uF, VCC_3V0, GND

44	VCC1	VLDO1	31 VCCA1V8_CODEC
45	VCC1	VLDO2	33 VCCA1V8_HDMI
23	VCC2	VLDO3	3 VCCA_1V8
24	VCC2	VLDO4	12 VCC_SDIO
59	VCC3	VLDO5	14 VCCA3V0_CODEC
60	VCC4	VLDO6	2 VCC_1V5
32	VCC6	VLDO7	5 VCCA0V9_HDMI
4	VCC7	VLDO8	15 VCC_3V0
8	VCC8		
13	VCC9	VLDOA	35 VCC3V3_S5
1	VCC10		
16	VCC11	VFB1	39 VFB1 R30 100R VDD_CENTER
10	VCC12	VFB2	29 VFB2 R31 100R VDD_CPU_L
		VFB3	56 VFB3
9	VCCRTC	VFB4	63 VFB4
36	VCCA		
		INT	49 PMIC_INT_L
34	VDC	SLEEP	50 PMIC_SLEEP_H
37	VREF	PWRON	51 POWER_ON
		BOOT0	52 BOOT0 R35 1K VCC_RTC
17	VDDIO	BOOT1	53 BOOT1 R36 1K
		DVSOK	54
47	NC	EXT_EN	55 EXT_EN
48	NC		

RK808-D

For DDR RAM voltage selection
DDR3 = R41->100K R42->200K = 1.5V
DDR3L = R41->43K R42->120K = 1.35V
LPDDR = R41->51K R42->200K = 1.25V

U9B

VCC3V3_S3
VPP_OTP 6, C49 0.1uF, C50 2.2uF 7 11, VCC_1V8, GND
OSC32KIN 65, OSC32KOUT 66, R42 10K RTC_CLKO_WIFI 67, R44 10K RTC_CLKO_SOC 68

	VPPOTP	SDA	18 I2C0_SDA_PMIC
	VSWOUT1	SCL	19 I2C0_SCL_PMIC
	VSWOUT2	NRESPWRON	20 RESET_L
		DVS2	21
	OSC32KIN	DVS1	22
	OSC32KOUT		
	CLK32KOUT2	REFGND	64
	CLK32KOUT1	AGND	30
		DGND	38
		GND1	40
	SW1	GND1	41
	SW1	GND2	27
	SW2	GND2	28
	SW2	GND3	57
	SW3	GND4	62
	SW4	GND5	46
		GND	69

VDD_CENTER SB1 VFB1 L6 1uH 42 43
VDD_CPU_L SB2 VFB2 L7 1uH 25 26
VCC_DDR SB3 VFB3 L8 2.2uH 58
VCC_1V8 SB4 VFB4 L9 2.2uH 61
RK808-D, GND

32.768KHz Crystal

X1, OSC32KIN, OSC32KOUT, C31 20pF, C32 20pF, GND, GND

Decoupling Cap

VCC3V3_SYS
VCC3V3_S5, C35 1uF, C36 1uF, C37 1uF, C38 1uF, C39 1uF, C40 1uF, C41 1uF, C42 1uF
POWER_ON C44 0.1uF
VPP_OTP C46 0.1uF
VCC_3V0, C47 1uF, C48 2.2uF, GND

Pull-Up Resistor

VCC_3V0
I2C0_SDA_PMIC R41 4.7K
I2C0_SCL_PMIC R43 4.7K

Decoupling Cap

VCC3V3_S3, VCC_1V8, C51 22uF, C52 22uF, C53 22uF, C54 22uF, GND
VCC_5V, C55 22uF, C56 22uF, C57 1uF, C58 1uF, C60 1uF, VCCA_1V8, C62 22uF, GND

Supporting Circuit

VCC3V3_S5, VDC_RK8 C59 22nF, C61 4.7uF, R47 2M, D7 B5819WS, R48 10K, GND, GND

Test Points

VCCA3V0_CODEC	P6
VCC_SDIO	P7
VCCA0V9_HDMI	P8
VCCA1V8_HDMI	P9
VCCA_1V8	P11
VCC_RTC	P12
VCC_3V0	P13
VCC3V3_SYS	P14
VCC3V3_S5	P15
VDD_CENTER	P17
VDD_CPU_L	P18
VDD_DDR	P19
VCC_1V8	P20
VCC_5V	P21

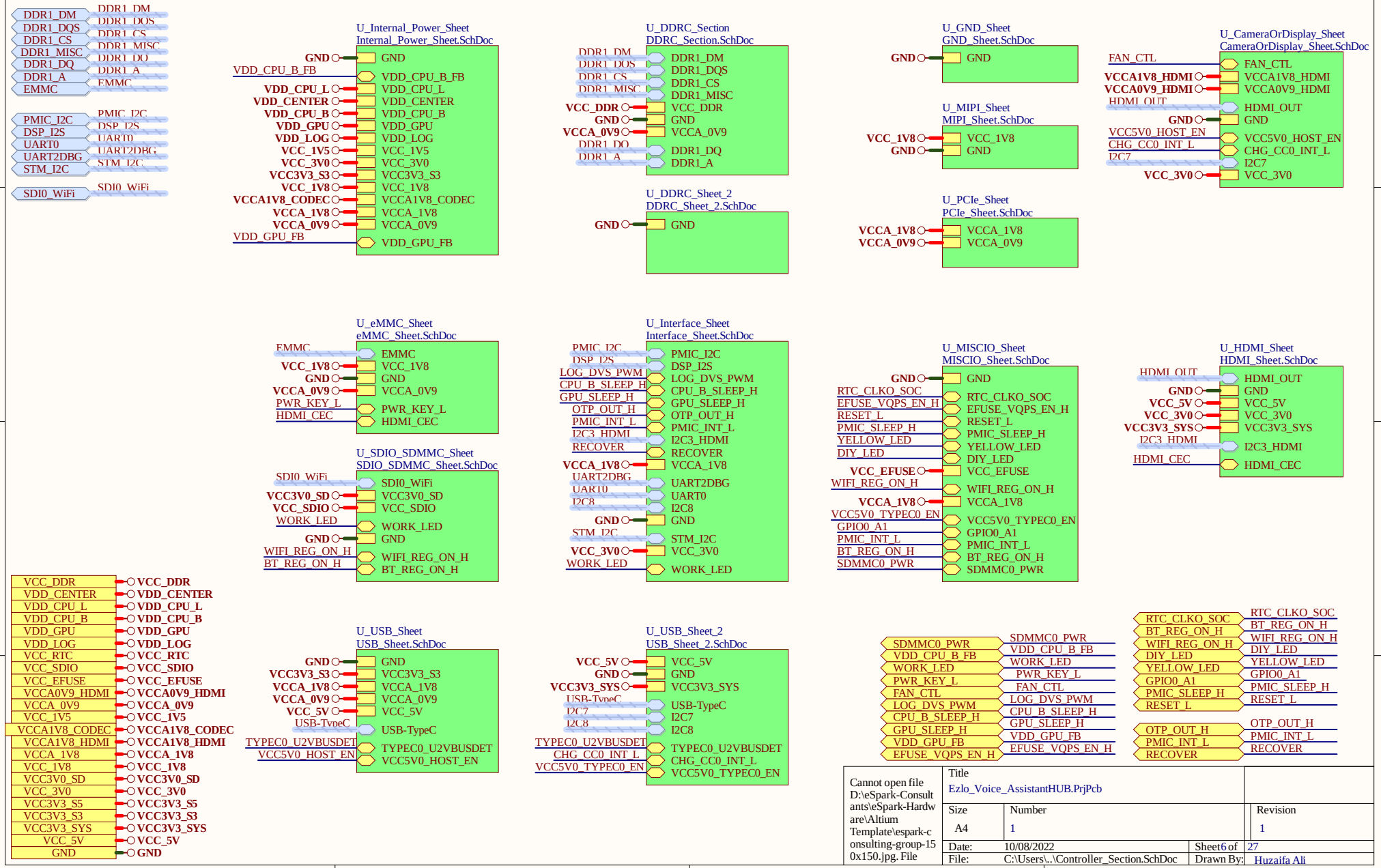
Boot Mode Selection

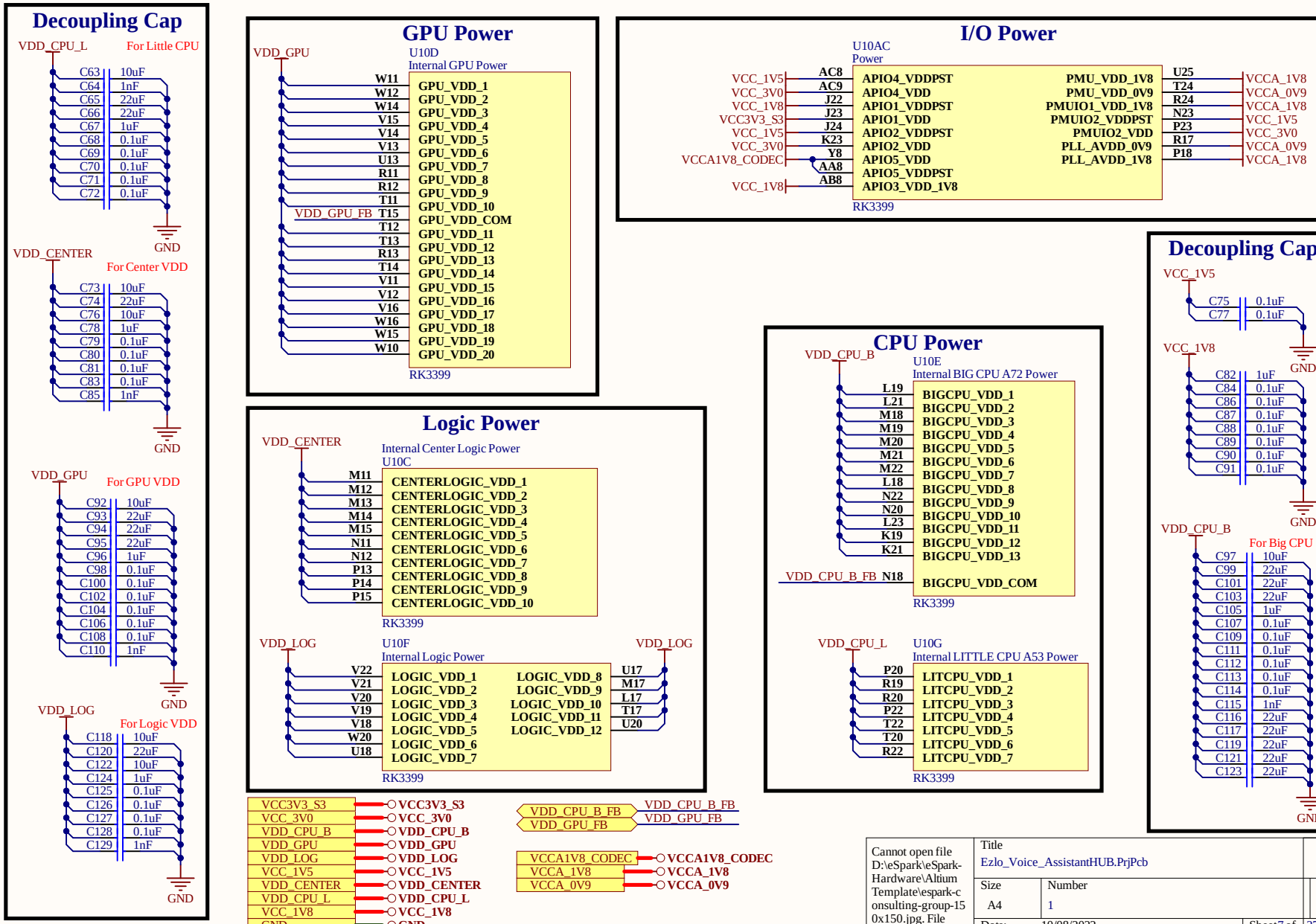
BOOT0, P10 1 2, 61300211121, GND
BOOT1, P16 1 2, 61300211121, GND

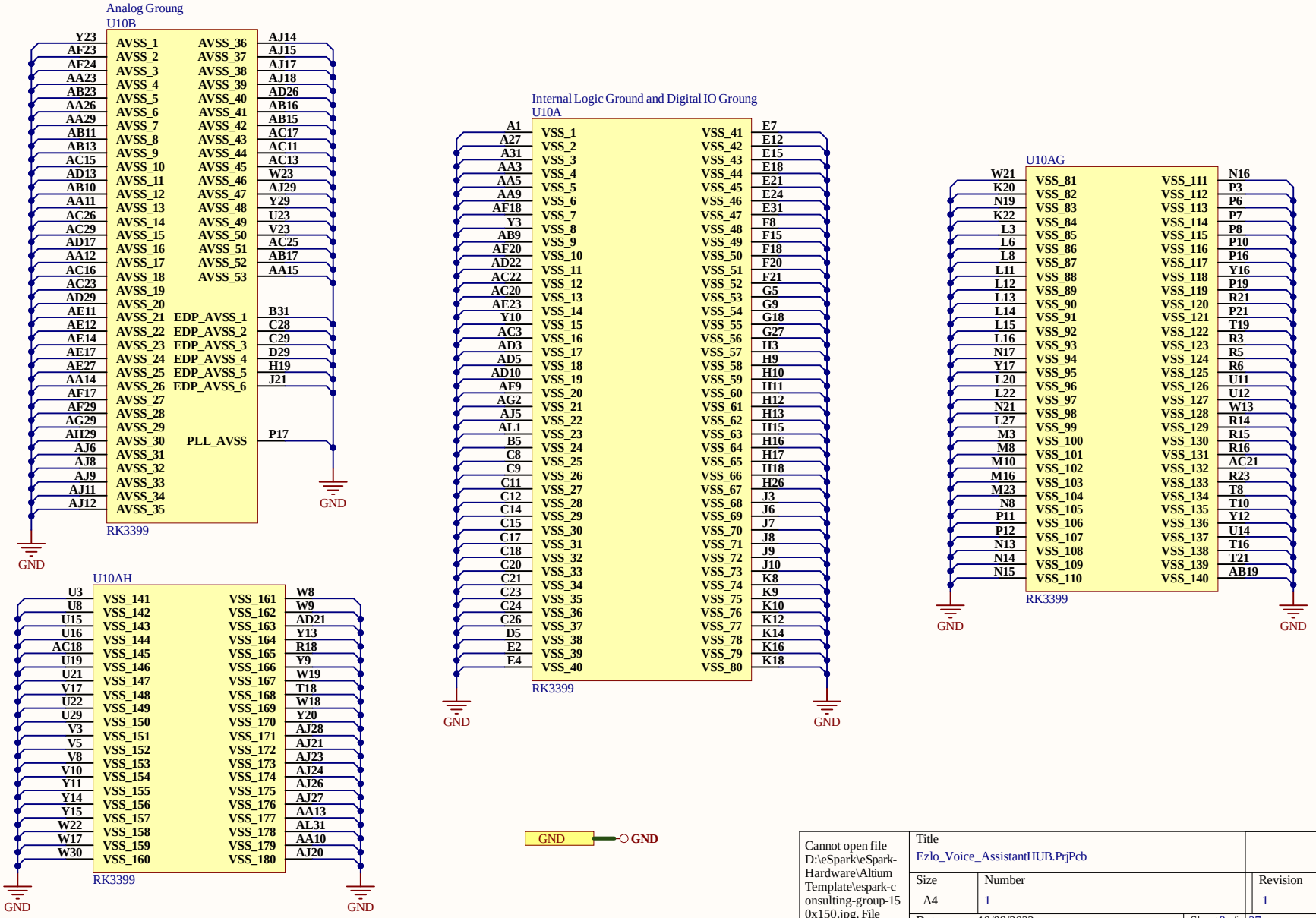
VCCA1V8_CODEC	VCCA1V8_CODEC	VCC_RTC	VCC_RTC
VCCA_1V8	VCCA_1V8	VCC_3V0	VCC_3V0
VCCA0V9_HDMI	VCCA0V9_HDMI	VCC_SDIO	VCC_SDIO
VCCA1V8_HDMI	VCCA1V8_HDMI	VCC3V3_S3	VCC3V3_S3
VCC3V3_SYS	VCC3V3_SYS	VCC_1V5	VCC_1V5
VCC3V3_S5	VCC3V3_S5		
VDD_CENTER	VDD_CENTER		
VDD_CPU_L	VDD_CPU_L		
VCC_DDR	VCC_DDR		
VCC_1V8	VCC_1V8		
VCC_5V	VCC_5V		
GND	GND		

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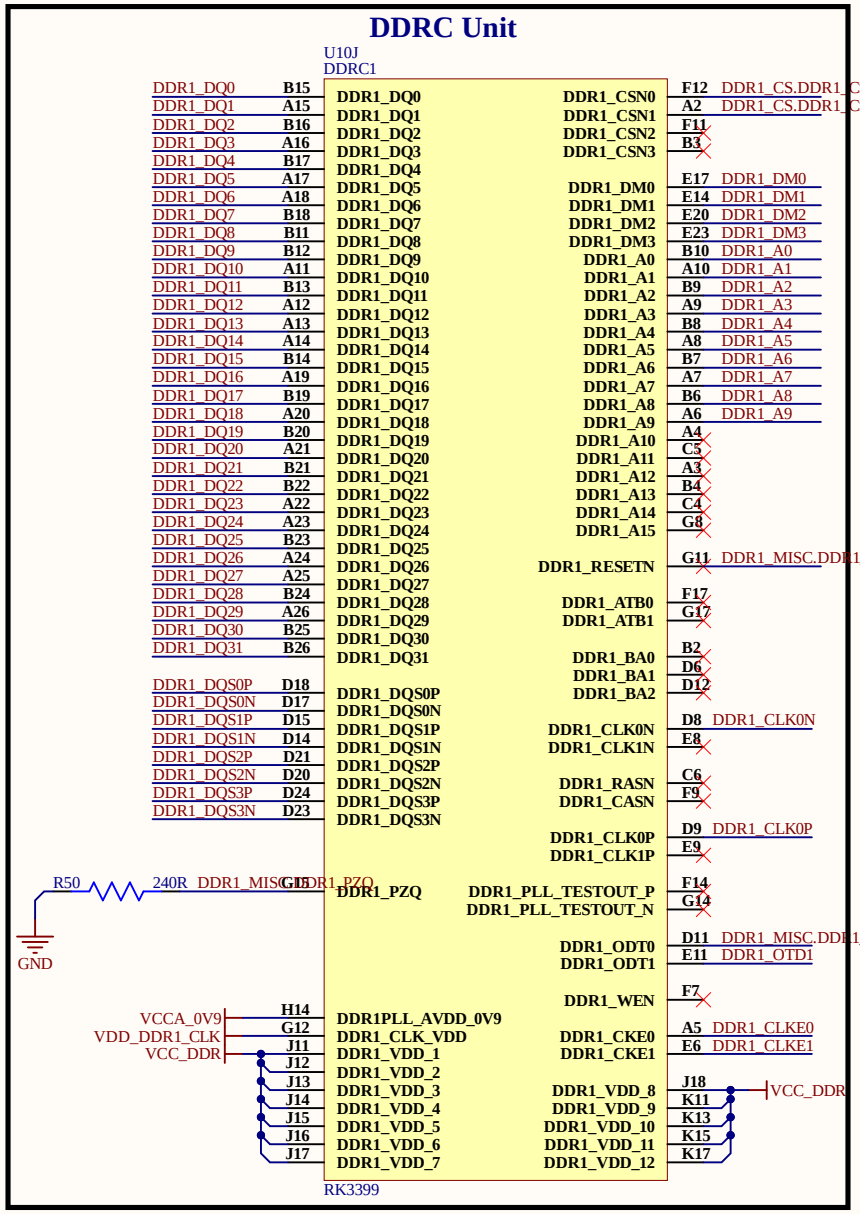
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		Revision	1
Date:	10/08/2022	Sheet 5 of	27
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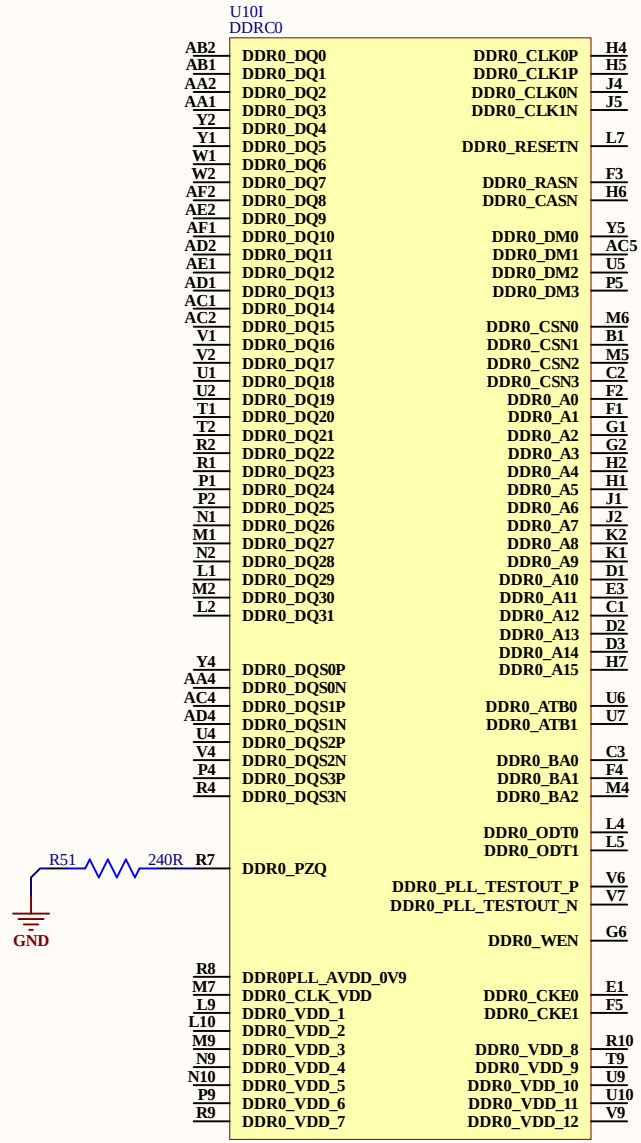

DDR1_DM
DDR1_DQS
DDR1_CS
DDR1_MISC
DDR1_DQ
DDR1_A
EMMC
PMIC_I2C
DSP_I2S
UART0
UART2DBG
STM_I2C
SDI0_WiFi
U_Internal_Power_Sheet
Internal_Power_Sheet.SchDoc
GND
VDD_CPU_B_FB
VDD_CPU_L
VDD_CENTER
VDD_CPU_B
VDD_GPU
VDD_LOG
VCC_1V5
VCC_3V0
VCC3V3_S3
VCC_1V8
VCCA1V8_CODEC
VCCA_1V8
VCCA_0V9
VDD_GPU_FB
U_DDRC_Section
DDRC_Section.SchDoc
VCC_DDR
U_DDRC_Sheet_2
DDRC_Sheet_2.SchDoc
U_GND_Sheet
GND_Sheet.SchDoc
U_MIPI_Sheet
MIPI_Sheet.SchDoc
U_PCIe_Sheet
PCIe_Sheet.SchDoc
U_CameraOrDisplay_Sheet
CameraOrDisplay_Sheet.SchDoc
FAN_CTL
VCCA1V8_HDMI
VCCA0V9_HDMI
HDMI_OUT
VCC5V0_HOST_EN
CHG_CC0_INT_L
I2C7
U_eMMC_Sheet
eMMC_Sheet.SchDoc
PWR_KEY_L
HDMI_CEC
U_SDIO_SDMMC_Sheet
SDIO_SDMMC_Sheet.SchDoc
WORK_LED
WIFI_REG_ON_H
BT_REG_ON_H
U_Interface_Sheet
Interface_Sheet.SchDoc
LOG_DVS_PWM
CPU_B_SLEEP_H
GPU_SLEEP_H
OTP_OUT_H
PMIC_INT_L
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UART2DBG
UART0
I2C8
STM_I2C
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MISCIO_Sheet.SchDoc
RTC_CLKO_SOC
EFUSE_VQPS_EN_H
RESET_L
PMIC_SLEEP_H
YELLOW_LED
DIY_LED
VCC_EFUSE
WIFI_REG_ON_H
VCC5V0_TYPEC0_EN
GPIO0_A1
SDMMC0_PWR
U_HDMI_Sheet
HDMI_Sheet.SchDoc
VCC_5V
VCC3V3_SYS
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USB_Sheet.SchDoc
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TYPEC0_U2VBUSDET
VCC5V0_HOST_EN
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USB_Sheet_2.SchDoc
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VCC_SDIO
VCCA0V9_HDMI
VCC_1V5
VCC3V0_SD
VCC3V3_S5
VCC3V3_S3
VDD_GPU
VDD_LOG
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Number
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Revision
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Date:
10/08/2022
Sheet 6 of 27
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Drawn By:
Huzaifa Ali

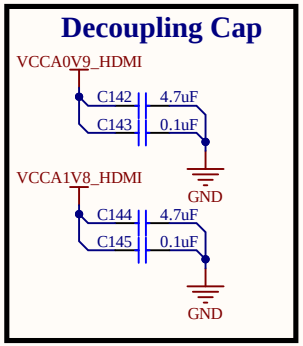

Decoupling Cap
VDD_CPU_L
For Little CPU
C63 10uF
C64 1nF
C65 22uF
C66 22uF
C67 1uF
C68 0.1uF
C69 0.1uF
C70 0.1uF
C71 0.1uF
C72 0.1uF
GND
VDD_CENTER
For Center VDD
C73 10uF
C74 22uF
C76 10uF
C78 1uF
C79 0.1uF
C80 0.1uF
C81 0.1uF
C83 0.1uF
C85 1nF
GND
VDD_GPU
For GPU VDD
C92 10uF
C93 22uF
C94 22uF
C95 22uF
C96 1uF
C98 0.1uF
C100 0.1uF
C102 0.1uF
C104 0.1uF
C106 0.1uF
C108 0.1uF
C110 1nF
GND
VDD_LOG
For Logic VDD
C118 10uF
C120 22uF
C122 10uF
C124 1uF
C125 0.1uF
C126 0.1uF
C127 0.1uF
C128 0.1uF
C129 1nF
GND
GPU Power
VDD_GPU
U10D
Internal GPU Power
W11 GPU_VDD_1
W12 GPU_VDD_2
W14 GPU_VDD_3
V15 GPU_VDD_4
V14 GPU_VDD_5
V13 GPU_VDD_6
U13 GPU_VDD_7
R11 GPU_VDD_8
R12 GPU_VDD_9
T11 GPU_VDD_10
VDD_GPU_FB T15 GPU_VDD_COM
T12 GPU_VDD_11
T13 GPU_VDD_12
R13 GPU_VDD_13
T14 GPU_VDD_14
V11 GPU_VDD_15
V12 GPU_VDD_16
V16 GPU_VDD_17
W16 GPU_VDD_18
W15 GPU_VDD_19
W10 GPU_VDD_20
RK3399
I/O Power
U10AC
Power
VCC_1V5 AC8
VCC_3V0 AC9
VCC_1V8 J22
VCC3V3_S3 J23
VCC_1V5 J24
VCC_3V0 K23
VCCA1V8_CODEC Y8
AA8
VCC_1V8 AB8
APIO4_VDDPST
APIO4_VDD
APIO1_VDDPST
APIO1_VDD
APIO2_VDDPST
APIO2_VDD
APIO5_VDD
APIO5_VDDPST
APIO3_VDD_1V8
PMU_VDD_1V8 U25
PMU_VDD_0V9 T24
PMUIO1_VDD_1V8 R24
PMUIO2_VDDPST N23
PMUIO2_VDD P23
PLL_AVDD_0V9 R17
PLL_AVDD_1V8 P18
VCCA_1V8
VCCA_0V9
VCCA_1V8
VCC_1V5
VCC_3V0
VCCA_0V9
VCCA_1V8
RK3399
Logic Power
VDD_CENTER
Internal Center Logic Power
U10C
M11 CENTERLOGIC_VDD_1
M12 CENTERLOGIC_VDD_2
M13 CENTERLOGIC_VDD_3
M14 CENTERLOGIC_VDD_4
M15 CENTERLOGIC_VDD_5
N11 CENTERLOGIC_VDD_6
N12 CENTERLOGIC_VDD_7
P13 CENTERLOGIC_VDD_8
P14 CENTERLOGIC_VDD_9
P15 CENTERLOGIC_VDD_10
RK3399
VDD_LOG
U10F
Internal Logic Power
V22 LOGIC_VDD_1
V21 LOGIC_VDD_2
V20 LOGIC_VDD_3
V19 LOGIC_VDD_4
V18 LOGIC_VDD_5
W20 LOGIC_VDD_6
U18 LOGIC_VDD_7
LOGIC_VDD_8 U17
LOGIC_VDD_9 M17
LOGIC_VDD_10 L17
LOGIC_VDD_11 T17
LOGIC_VDD_12 U20
VDD_LOG
RK3399
CPU Power
VDD_CPU_B
U10E
Internal BIG CPU A72 Power
L19 BIGCPU_VDD_1
L21 BIGCPU_VDD_2
M18 BIGCPU_VDD_3
M19 BIGCPU_VDD_4
M20 BIGCPU_VDD_5
M21 BIGCPU_VDD_6
M22 BIGCPU_VDD_7
L18 BIGCPU_VDD_8
N22 BIGCPU_VDD_9
N20 BIGCPU_VDD_10
L23 BIGCPU_VDD_11
K19 BIGCPU_VDD_12
K21 BIGCPU_VDD_13
VDD_CPU_B_FB N18 BIGCPU_VDD_COM
RK3399
VDD_CPU_L
U10G
Internal LITTLE CPU A53 Power
P20 LITCPU_VDD_1
R19 LITCPU_VDD_2
R20 LITCPU_VDD_3
P22 LITCPU_VDD_4
T22 LITCPU_VDD_5
T20 LITCPU_VDD_6
R22 LITCPU_VDD_7
RK3399
Decoupling Cap
VCC_1V5
C75 0.1uF
C77 0.1uF
GND
VCC_1V8
C82 1uF
C84 0.1uF
C86 0.1uF
C87 0.1uF
C88 0.1uF
C89 0.1uF
C90 0.1uF
C91 0.1uF
GND
VDD_CPU_B
For Big CPU
C97 10uF
C99 22uF
C101 22uF
C103 22uF
C105 1uF
C107 0.1uF
C109 0.1uF
C111 0.1uF
C112 0.1uF
C113 0.1uF
C114 0.1uF
C115 1nF
C116 22uF
C117 22uF
C119 22uF
C121 22uF
C123 22uF
GND
VCC3V3_S3
VCC_3V0
VDD_CPU_B
VDD_GPU
VDD_LOG
VCC_1V5
VDD_CENTER
VDD_CPU_L
VCC_1V8
GND
VDD_CPU_B_FB
VDD_GPU_FB
VCCA1V8_CODEC
VCCA_1V8
VCCA_0V9
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size A4
Number 1
Revision 1
Date: 10/08/2022
Sheet 7 of 27
File: C:\Users\..\Internal_Power_Sheet.SchDoc
Drawn By: Huzaifa Ali

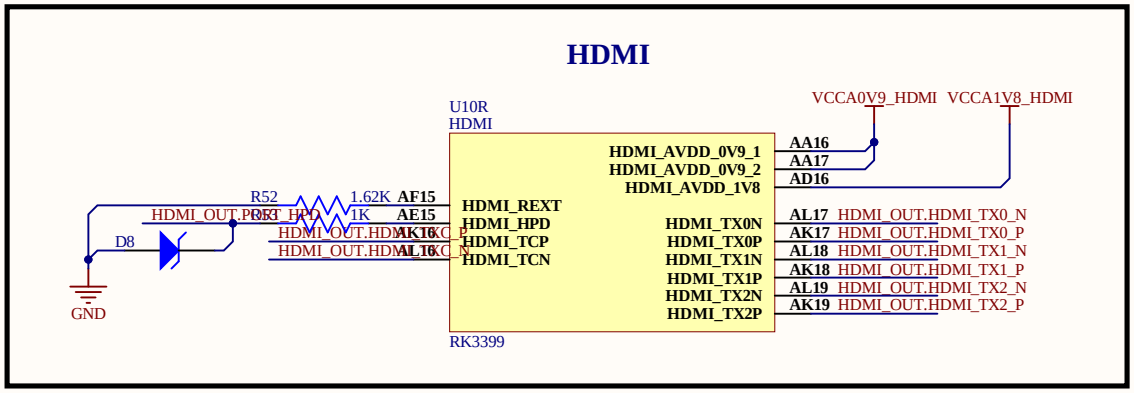


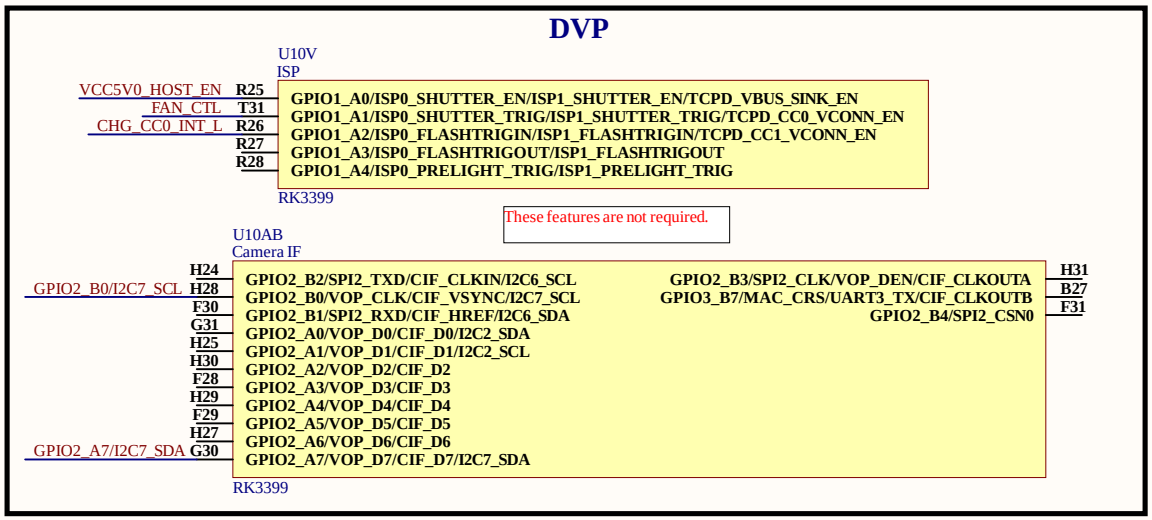
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	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 8 of 27	
	File: C:\Users\..\GND_Sheet.SchDoc	Drawn By: Huzaifa Ali	

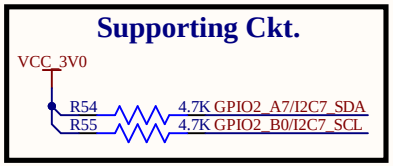

DDRC Unit
U10J
DDRC1
RK3399
DDR1_DQ0 B15
DDR1_DQ1 A15
DDR1_DQ2 B16
DDR1_DQ3 A16
DDR1_DQ4 B17
DDR1_DQ5 A17
DDR1_DQ6 A18
DDR1_DQ7 B18
DDR1_DQ8 B11
DDR1_DQ9 B12
DDR1_DQ10 A11
DDR1_DQ11 B13
DDR1_DQ12 A12
DDR1_DQ13 A13
DDR1_DQ14 A14
DDR1_DQ15 B14
DDR1_DQ16 A19
DDR1_DQ17 B19
DDR1_DQ18 A20
DDR1_DQ19 B20
DDR1_DQ20 A21
DDR1_DQ21 B21
DDR1_DQ22 B22
DDR1_DQ23 A22
DDR1_DQ24 A23
DDR1_DQ25 B23
DDR1_DQ26 A24
DDR1_DQ27 A25
DDR1_DQ28 B24
DDR1_DQ29 A26
DDR1_DQ30 B25
DDR1_DQ31 B26
DDR1_DQS0P D18
DDR1_DQS0N D17
DDR1_DQS1P D15
DDR1_DQS1N D14
DDR1_DQS2P D21
DDR1_DQS2N D20
DDR1_DQS3P D24
DDR1_DQS3N D23
R50 240R DDR1_MISC.DDR1_PZQ
GND
VCCA_0V9 H14
VDD_DDR1_CLK G12
VCC_DDR J11
J12
J13
J14
J15
J16
J17
DDR1_DQ0 DDR1_DQ1 DDR1_DQ2 DDR1_DQ3 DDR1_DQ4 DDR1_DQ5 DDR1_DQ6 DDR1_DQ7 DDR1_DQ8 DDR1_DQ9 DDR1_DQ10 DDR1_DQ11 DDR1_DQ12 DDR1_DQ13 DDR1_DQ14 DDR1_DQ15 DDR1_DQ16 DDR1_DQ17 DDR1_DQ18 DDR1_DQ19 DDR1_DQ20 DDR1_DQ21 DDR1_DQ22 DDR1_DQ23 DDR1_DQ24 DDR1_DQ25 DDR1_DQ26 DDR1_DQ27 DDR1_DQ28 DDR1_DQ29 DDR1_DQ30 DDR1_DQ31
DDR1_DQS0P DDR1_DQS0N DDR1_DQS1P DDR1_DQS1N DDR1_DQS2P DDR1_DQS2N DDR1_DQS3P DDR1_DQS3N
DDR1_PZQ
DDR1PLL_AVDD_0V9 DDR1_CLK_VDD DDR1_VDD_1 DDR1_VDD_2 DDR1_VDD_3 DDR1_VDD_4 DDR1_VDD_5 DDR1_VDD_6 DDR1_VDD_7
DDR1_CSN0 F12 DDR1_CS.DDR1_CSN0
DDR1_CSN1 A2 DDR1_CS.DDR1_CSN1
DDR1_CSN2 F11
DDR1_CSN3 B3
DDR1_DM0 E17 DDR1_DM0
DDR1_DM1 E14 DDR1_DM1
DDR1_DM2 E20 DDR1_DM2
DDR1_DM3 E23 DDR1_DM3
DDR1_A0 B10 DDR1_A0
DDR1_A1 A10 DDR1_A1
DDR1_A2 B9 DDR1_A2
DDR1_A3 A9 DDR1_A3
DDR1_A4 B8 DDR1_A4
DDR1_A5 A8 DDR1_A5
DDR1_A6 B7 DDR1_A6
DDR1_A7 A7 DDR1_A7
DDR1_A8 B6 DDR1_A8
DDR1_A9 A6 DDR1_A9
DDR1_A10 A4
DDR1_A11 C5
DDR1_A12 A3
DDR1_A13 B4
DDR1_A14 C4
DDR1_A15 G8
DDR1_RESETN G11 DDR1_MISC.DDR1_RST
DDR1_ATB0 F17
DDR1_ATB1 G17
DDR1_BA0 B2
DDR1_BA1 D6
DDR1_BA2 D12
DDR1_CLK0N D8 DDR1_CLK0N
DDR1_CLK1N E8
DDR1_RASN C6
DDR1_CASN F9
DDR1_CLK0P D9 DDR1_CLK0P
DDR1_CLK1P E9
DDR1_PLL_TESTOUT_P F14
DDR1_PLL_TESTOUT_N G14
DDR1_ODT0 D11 DDR1_MISC.DDR1_OTD
DDR1_ODT1 E11 DDR1_OTD1
DDR1_WEN F7
DDR1_CKE0 A5 DDR1_CKE0
DDR1_CKE1 E6 DDR1_CKE1
DDR1_VDD_8 J18
DDR1_VDD_9 K11
DDR1_VDD_10 K13
DDR1_VDD_11 K15
DDR1_VDD_12 K17
VCC_DDR
DDR1_A
DDR1_A0 DDR1_A1 DDR1_A2 DDR1_A3 DDR1_A4 DDR1_A5 DDR1_A6 DDR1_A7 DDR1_A8 DDR1_A9
DDR1_DQ
DDR1_DQ0 DDR1_DQ1 DDR1_DQ2 DDR1_DQ3 DDR1_DQ4 DDR1_DQ5 DDR1_DQ6 DDR1_DQ7 DDR1_DQ8 DDR1_DQ9 DDR1_DQ10 DDR1_DQ11 DDR1_DQ12 DDR1_DQ13 DDR1_DQ14 DDR1_DQ15 DDR1_DQ16 DDR1_DQ17 DDR1_DQ18 DDR1_DQ19 DDR1_DQ20 DDR1_DQ21 DDR1_DQ22 DDR1_DQ23 DDR1_DQ24 DDR1_DQ25 DDR1_DQ26 DDR1_DQ27 DDR1_DQ28 DDR1_DQ29 DDR1_DQ30 DDR1_DQ31
Test Points
DDR1_MISC.DDR1_OTD P22
DDR1_OTD1 P23
VCCA_0V9
VCC_DDR
GND
Decoupling Cap
VCC_DDR VCCA_0V9
C130 0.1uF
C131 0.1uF
C132 0.1uF
C133 0.1uF
C134 0.1uF
C135 0.1uF
C136 0.1uF
C137 0.1uF
C138 0.1uF
C139 0.1uF
C140 0.1uF
GND
VDD_DDR1_CLK
R49 0R VCC_DDR
C141 0.1uF GND
DDR1_DQS
DDR1_DQS0P DDR1_DQS0N DDR1_DQS1P DDR1_DQS1N DDR1_DQS2P DDR1_DQS2N DDR1_DQS3P DDR1_DQS3N
DDR1_DM
DDR1_DM0 DDR1_DM1 DDR1_DM2 DDR1_DM3
DDR1_CS
DDR1_CSN0 DDR1_CS.DDR1_CSN0
DDR1_CSN1 DDR1_CS.DDR1_CSN1
DDR1_MISC
DDR1_CLK0N DDR1_CLK0P DDR1_PZQ DDR1_MISC.DDR1_PZQ DDR1_OTD DDR1_MISC.DDR1_OTD DDR1_CLKE0 DDR1_CLKE1 DDR1_RST DDR1_MISC.DDR1_RST
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size A4
Number 1
Revision 1
Date: 10/08/2022
Sheet 9 of 27
File: C:\Users\..\DDRC_Section.SchDoc
Drawn By: Huzaifa Ali

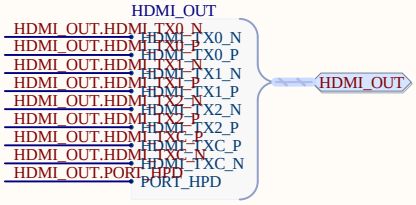

U10I
DDRC0
AB2 DDR0_DQ0
AB1 DDR0_DQ1
AA2 DDR0_DQ2
AA1 DDR0_DQ3
Y2 DDR0_DQ4
Y1 DDR0_DQ5
W1 DDR0_DQ6
W2 DDR0_DQ7
AF2 DDR0_DQ8
AE2 DDR0_DQ9
AF1 DDR0_DQ10
AD2 DDR0_DQ11
AE1 DDR0_DQ12
AD1 DDR0_DQ13
AC1 DDR0_DQ14
AC2 DDR0_DQ15
V1 DDR0_DQ16
V2 DDR0_DQ17
U1 DDR0_DQ18
U2 DDR0_DQ19
T1 DDR0_DQ20
T2 DDR0_DQ21
R2 DDR0_DQ22
R1 DDR0_DQ23
P1 DDR0_DQ24
P2 DDR0_DQ25
N1 DDR0_DQ26
M1 DDR0_DQ27
N2 DDR0_DQ28
L1 DDR0_DQ29
M2 DDR0_DQ30
L2 DDR0_DQ31
Y4 DDR0_DQS0P
AA4 DDR0_DQS0N
AC4 DDR0_DQS1P
AD4 DDR0_DQS1N
U4 DDR0_DQS2P
V4 DDR0_DQS2N
P4 DDR0_DQS3P
R4 DDR0_DQS3N
R51 240R R7
DDR0_PZQ
R8 DDR0PLL_AVDD_0V9
M7 DDR0_CLK_VDD
L9 DDR0_VDD_1
L10 DDR0_VDD_2
M9 DDR0_VDD_3
N9 DDR0_VDD_4
N10 DDR0_VDD_5
P9 DDR0_VDD_6
R9 DDR0_VDD_7
DDR0_CLK0P H4
DDR0_CLK1P H5
DDR0_CLK0N J4
DDR0_CLK1N J5
DDR0_RESETN L7
DDR0_RASN F3
DDR0_CASN H6
DDR0_DM0 Y5
DDR0_DM1 AC5
DDR0_DM2 U5
DDR0_DM3 P5
DDR0_CSN0 M6
DDR0_CSN1 B1
DDR0_CSN2 M5
DDR0_CSN3 C2
DDR0_A0 F2
DDR0_A1 F1
DDR0_A2 G1
DDR0_A3 G2
DDR0_A4 H2
DDR0_A5 H1
DDR0_A6 J1
DDR0_A7 J2
DDR0_A8 K2
DDR0_A9 K1
DDR0_A10 D1
DDR0_A11 E3
DDR0_A12 C1
DDR0_A13 D2
DDR0_A14 D3
DDR0_A15 H7
DDR0_ATB0 U6
DDR0_ATB1 U7
DDR0_BA0 C3
DDR0_BA1 F4
DDR0_BA2 M4
DDR0_ODT0 L4
DDR0_ODT1 L5
DDR0_PLL_TESTOUT_P V6
DDR0_PLL_TESTOUT_N V7
DDR0_WEN G6
DDR0_CKE0 E1
DDR0_CKE1 F5
DDR0_VDD_8 R10
DDR0_VDD_9 T9
DDR0_VDD_10 U9
DDR0_VDD_11 U10
DDR0_VDD_12 V9
RK3399
GND
These features are not required.
GND
GND
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size A4
Number 1
Revision 1
Date: 10/08/2022
Sheet 10 of 27
File: C:\Users\..\DDRC_Sheet_2.SchDoc
Drawn By: Huzaifa Ali





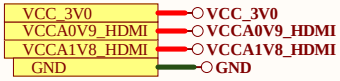






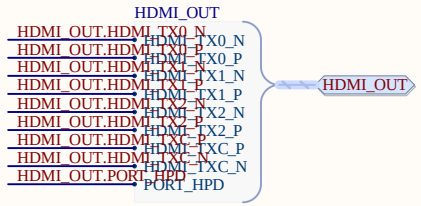


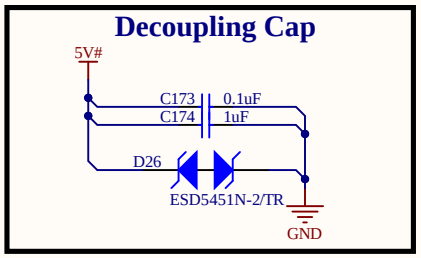


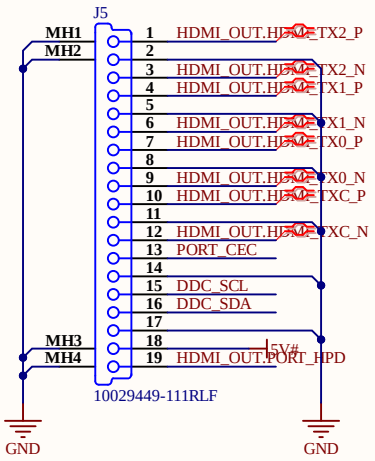


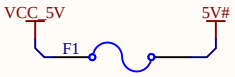
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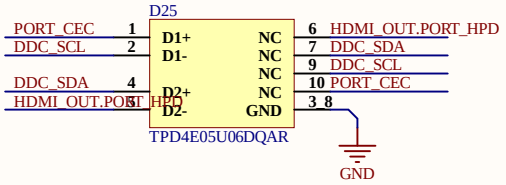
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Size	Number		Revision
A4	1		1
Date:	10/08/2022	Sheet 1 of	27
File:	C:\Users\..\CameraOrDisplay_Sheet.SchDoc	Drawn By:	Huzaifa Ali

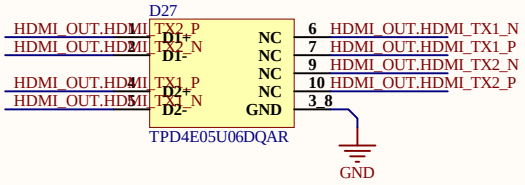


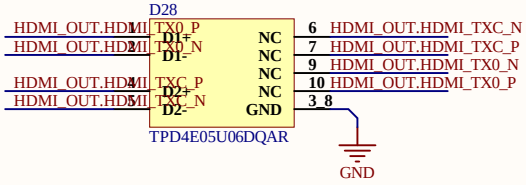


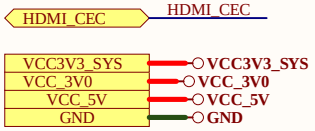




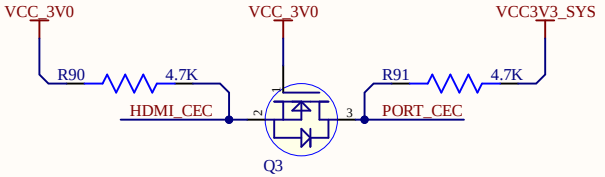


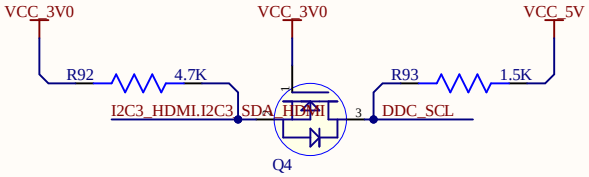


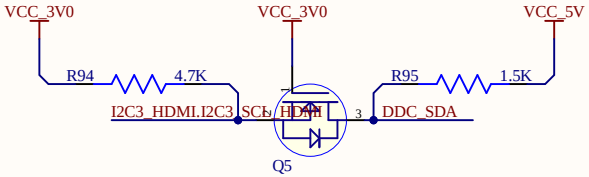




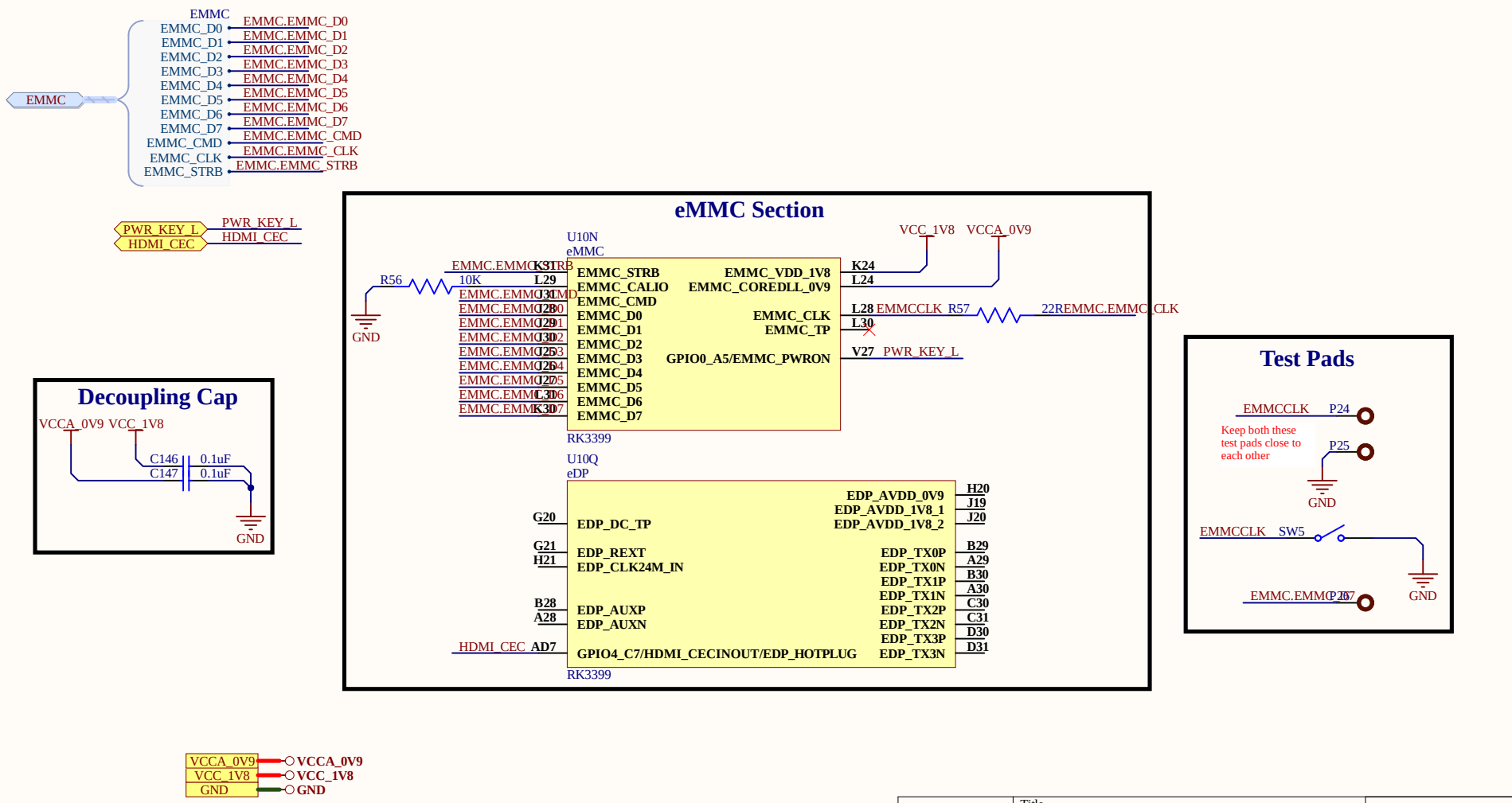




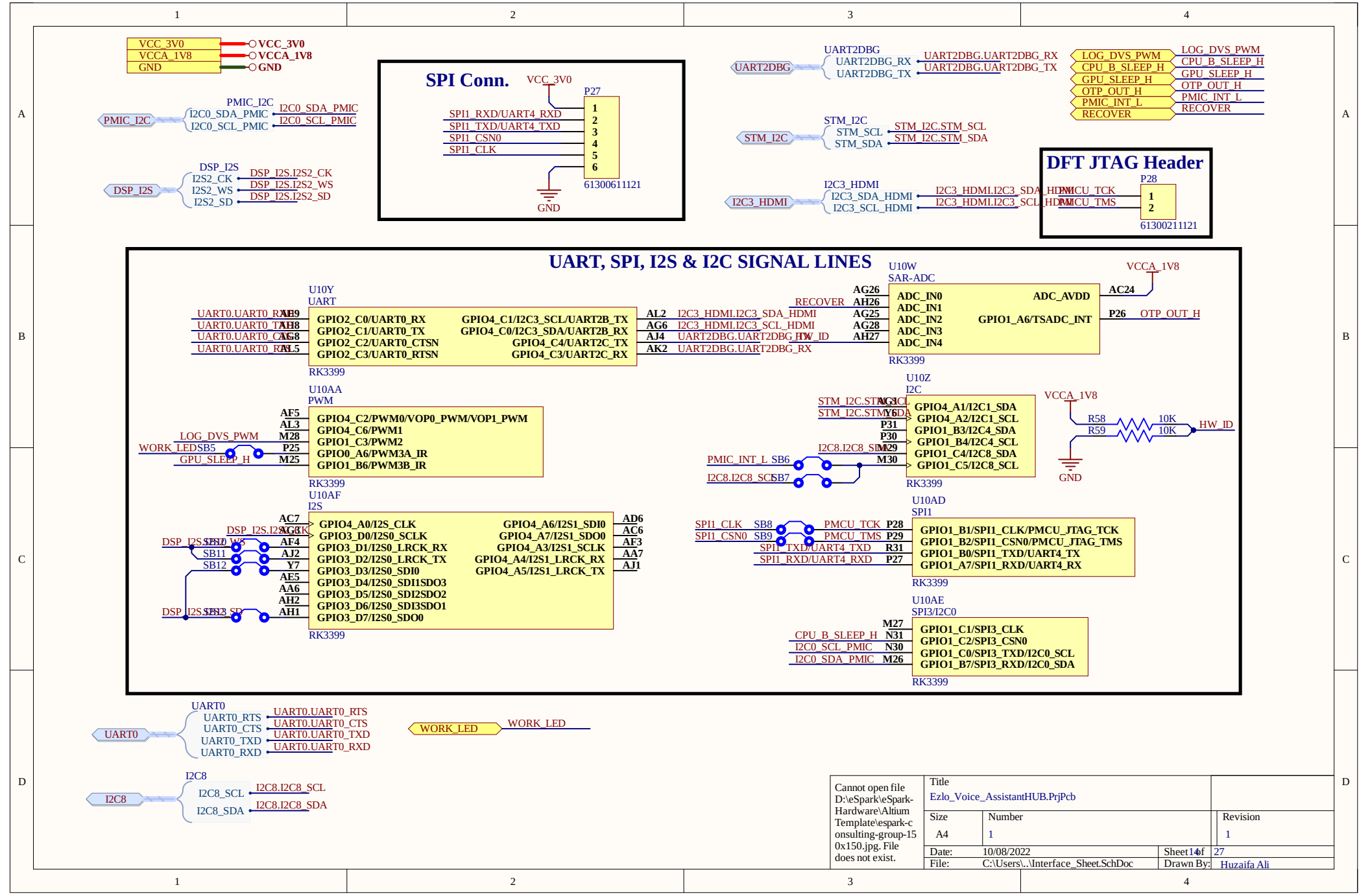


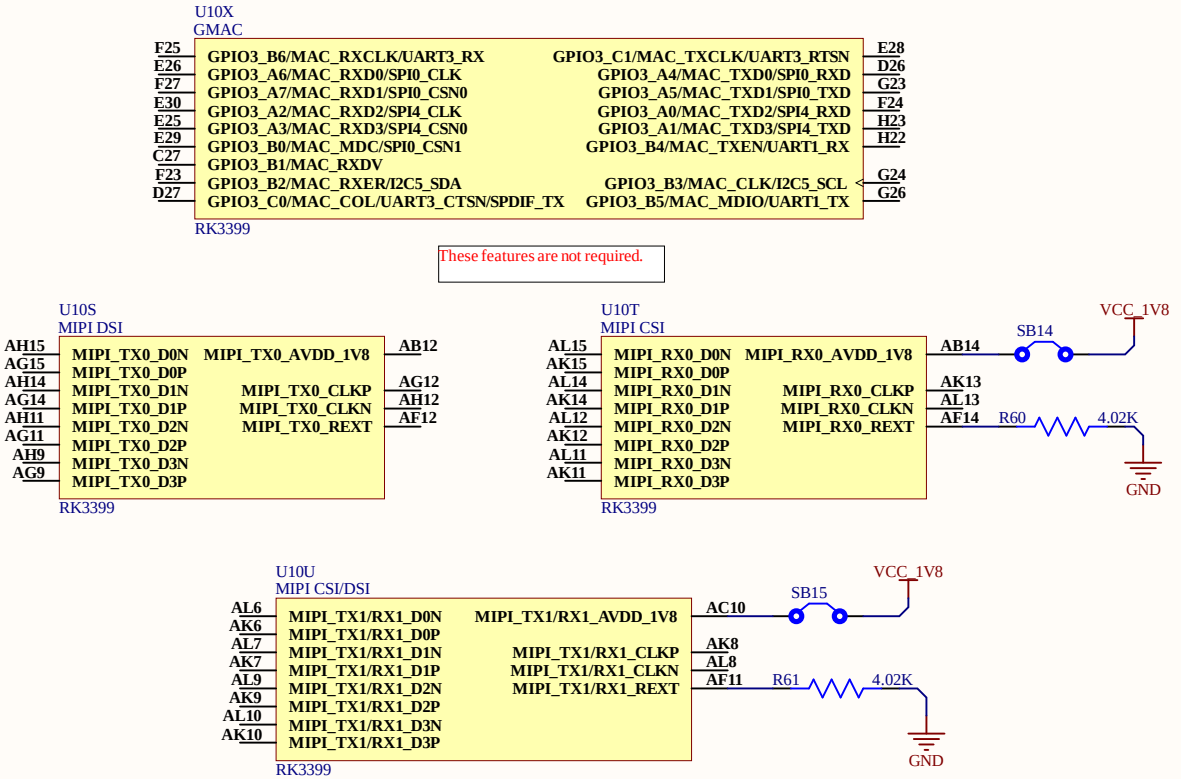


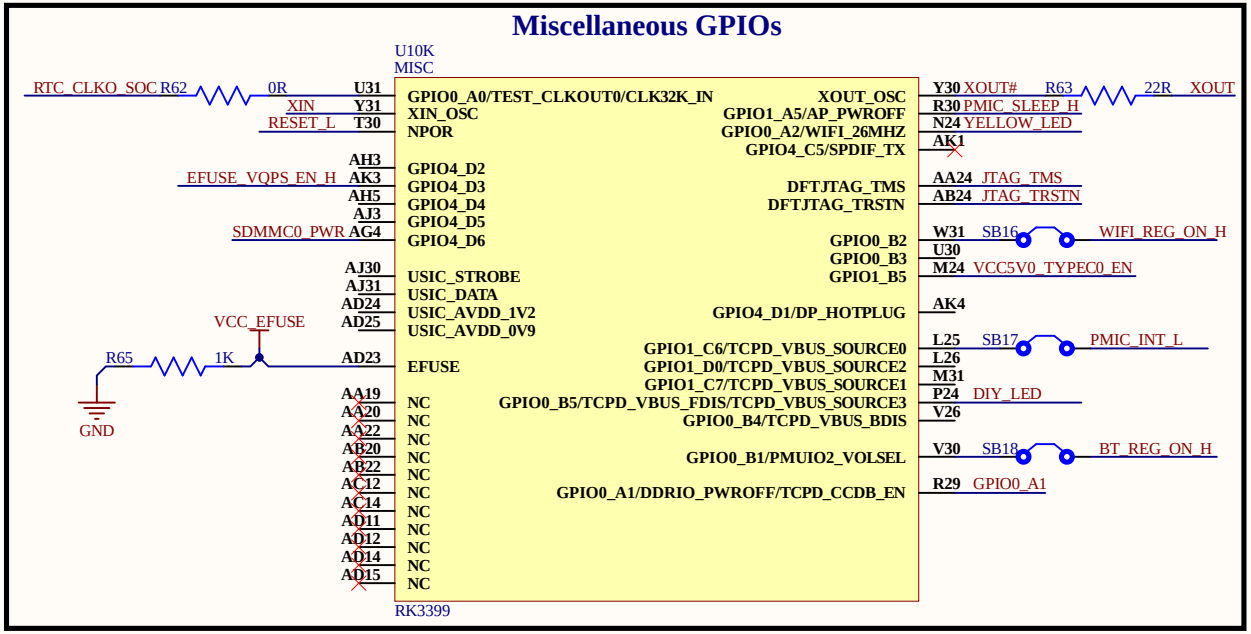
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File	Title: Ezlo_Voice_AssistantHUB.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 12 of 27	
	File: C:\Users\..\HDMI_Sheet.SchDoc	Drawn By: Huzaifa Ali	



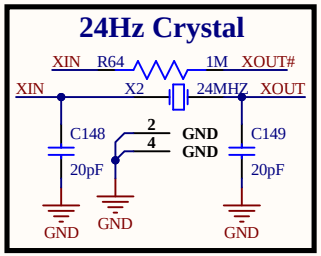
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_AssistantHUB.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 13 of 27	
	File: C:\Users\..\eMMC_Sheet.SchDoc	Drawn By: Huzaifa Ali	

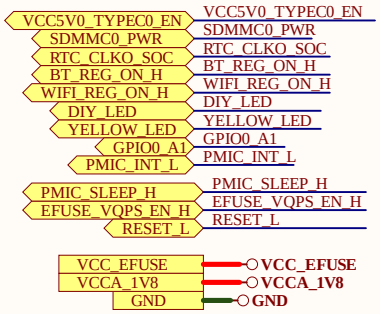

VCC_3V0
VCCA_1V8
GND
PMIC_I2C
I2C0_SDA_PMIC
I2C0_SCL_PMIC
DSP_I2S
I2S2_CK
I2S2_WS
I2S2_SD
SPI Conn.
SPI1_RXD/UART4_RXD
SPI1_TXD/UART4_TXD
SPI1_CSN0
SPI1_CLK
P27
61300611121
UART2DBG
UART2DBG_RX
UART2DBG_TX
LOG_DVS_PWM
CPU_B_SLEEP_H
GPU_SLEEP_H
OTP_OUT_H
PMIC_INT_L
RECOVER
STM_I2C
STM_SCL
STM_SDA
I2C3_HDMI
I2C3_SDA_HDMI
I2C3_SCL_HDMI
DFT JTAG Header
P28
61300211121
UART, SPI, I2S & I2C SIGNAL LINES
U10Y
UART
GPIO2_C0/UART0_RX
GPIO2_C1/UART0_TX
GPIO2_C2/UART0_CTSN
GPIO2_C3/UART0_RTSN
GPIO4_C1/I2C3_SCL/UART2B_TX
GPIO4_C0/I2C3_SDA/UART2B_RX
GPIO4_C4/UART2C_TX
GPIO4_C3/UART2C_RX
RK3399
U10W
SAR-ADC
ADC_IN0
ADC_IN1
ADC_IN2
ADC_IN3
ADC_IN4
ADC_AVDD
GPIO1_A6/TSADC_INT
U10AA
PWM
GPIO4_C2/PWM0/VOP0_PWM/VOP1_PWM
GPIO4_C6/PWM1
GPIO1_C3/PWM2
GPIO0_A6/PWM3A_IR
GPIO1_B6/PWM3B_IR
U10Z
I2C
GPIO4_A1/I2C1_SDA
GPIO4_A2/I2C1_SCL
GPIO1_B3/I2C4_SDA
GPIO1_B4/I2C4_SCL
GPIO1_C4/I2C8_SDA
GPIO1_C5/I2C8_SCL
R58
R59
10K
HW_ID
U10AF
I2S
GPIO4_A0/I2S_CLK
GPIO3_D0/I2S0_SCLK
GPIO3_D1/I2S0_LRCK_RX
GPIO3_D2/I2S0_LRCK_TX
GPIO3_D3/I2S0_SDI0
GPIO3_D4/I2S0_SDI1SDO3
GPIO3_D5/I2S0_SDI2SDO2
GPIO3_D6/I2S0_SDI3SDO1
GPIO3_D7/I2S0_SDO0
GPIO4_A6/I2S1_SDI0
GPIO4_A7/I2S1_SDO0
GPIO4_A3/I2S1_SCLK
GPIO4_A4/I2S1_LRCK_RX
GPIO4_A5/I2S1_LRCK_TX
U10AD
SPI1
GPIO1_B1/SPI1_CLK/PMCU_JTAG_TCK
GPIO1_B2/SPI1_CSN0/PMCU_JTAG_TMS
GPIO1_B0/SPI1_TXD/UART4_TX
GPIO1_A7/SPI1_RXD/UART4_RX
U10AE
SPI3/I2C0
GPIO1_C1/SPI3_CLK
GPIO1_C2/SPI3_CSN0
GPIO1_C0/SPI3_TXD/I2C0_SCL
GPIO1_B7/SPI3_RXD/I2C0_SDA
UART0
UART0_RTS
UART0_CTS
UART0_TXD
UART0_RXD
WORK_LED
I2C8
I2C8_SCL
I2C8_SDA
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.
Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size
A4
Number
1
Revision
1
Date:
10/08/2022
Sheet 14 of 27
File:
C:\Users\..\Interface_Sheet.SchDoc
Drawn By:
Huzaifa Ali

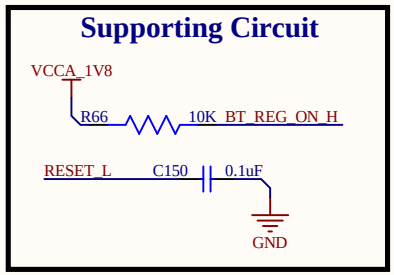

U10X
GMAC
F25 GPIO3_B6/MAC_RXCLK/UART3_RX
E26 GPIO3_A6/MAC_RXD0/SPI0_CLK
F27 GPIO3_A7/MAC_RXD1/SPI0_CSN0
E30 GPIO3_A2/MAC_RXD2/SPI4_CLK
E25 GPIO3_A3/MAC_RXD3/SPI4_CSN0
E29 GPIO3_B0/MAC_MDC/SPI0_CSN1
C27 GPIO3_B1/MAC_RXDV
F23 GPIO3_B2/MAC_RXER/I2C5_SDA
D27 GPIO3_C0/MAC_COL/UART3_CTSN/SPDIF_TX
GPIO3_C1/MAC_TXCLK/UART3_RTSN E28
GPIO3_A4/MAC_TXD0/SPI0_RXD D26
GPIO3_A5/MAC_TXD1/SPI0_TXD G23
GPIO3_A0/MAC_TXD2/SPI4_RXD F24
GPIO3_A1/MAC_TXD3/SPI4_TXD H23
GPIO3_B4/MAC_TXEN/UART1_RX H22
GPIO3_B3/MAC_CLK/I2C5_SCL G24
GPIO3_B5/MAC_MDIO/UART1_TX G26
RK3399
These features are not required.
U10S
MIPI DSI
AH15 MIPI_TX0_D0N
AG15 MIPI_TX0_D0P
AH14 MIPI_TX0_D1N
AG14 MIPI_TX0_D1P
AH11 MIPI_TX0_D2N
AG11 MIPI_TX0_D2P
AH9 MIPI_TX0_D3N
AG9 MIPI_TX0_D3P
MIPI_TX0_AVDD_1V8 AB12
MIPI_TX0_CLKP AG12
MIPI_TX0_CLKN AH12
MIPI_TX0_REXT AF12
RK3399
U10T
MIPI CSI
AL15 MIPI_RX0_D0N
AK15 MIPI_RX0_D0P
AL14 MIPI_RX0_D1N
AK14 MIPI_RX0_D1P
AL12 MIPI_RX0_D2N
AK12 MIPI_RX0_D2P
AL11 MIPI_RX0_D3N
AK11 MIPI_RX0_D3P
MIPI_RX0_AVDD_1V8 AB14
MIPI_RX0_CLKP AK13
MIPI_RX0_CLKN AL13
MIPI_RX0_REXT AF14
VCC_1V8
SB14
R60
4.02K
GND
RK3399
U10U
MIPI CSI/DSI
AL6 MIPI_TX1/RX1_D0N
AK6 MIPI_TX1/RX1_D0P
AL7 MIPI_TX1/RX1_D1N
AK7 MIPI_TX1/RX1_D1P
AL9 MIPI_TX1/RX1_D2N
AK9 MIPI_TX1/RX1_D2P
AL10 MIPI_TX1/RX1_D3N
AK10 MIPI_TX1/RX1_D3P
MIPI_TX1/RX1_AVDD_1V8 AC10
AK8
AL8
MIPI_TX1/RX1_CLKP
MIPI_TX1/RX1_CLKN
MIPI_TX1/RX1_REXT AF11
VCC_1V8
SB15
R61
4.02K
GND
RK3399
VCC_1V8
GND
VCC_1V8
GND
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.
Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size A4
Number 1
Revision 1
Date: 10/08/2022
Sheet 15 of 27
File: C:\Users\..\MIPI_Sheet.SchDoc
Drawn By: Huzaifa Ali

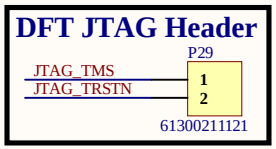


- AP means CPU core in RK3399 including Cortex A72 and Cortex A53.
- MCU and PMCU means built-in micro-controller in RK3399 core domain.



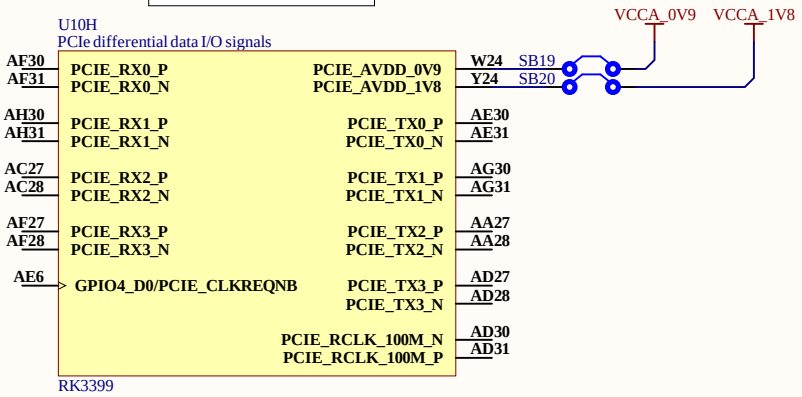






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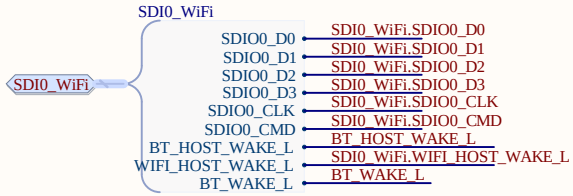
Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	10/08/2022	Sheet 16 of	27
File:	C:\Users\..\MISCIO_Sheet.SchDoc	Drawn By:	Huzaifa Ali

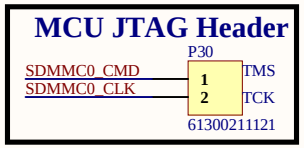


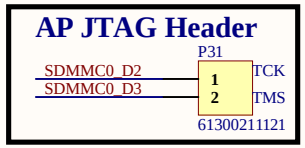
VCCA_1V8 — VCCA_1V8

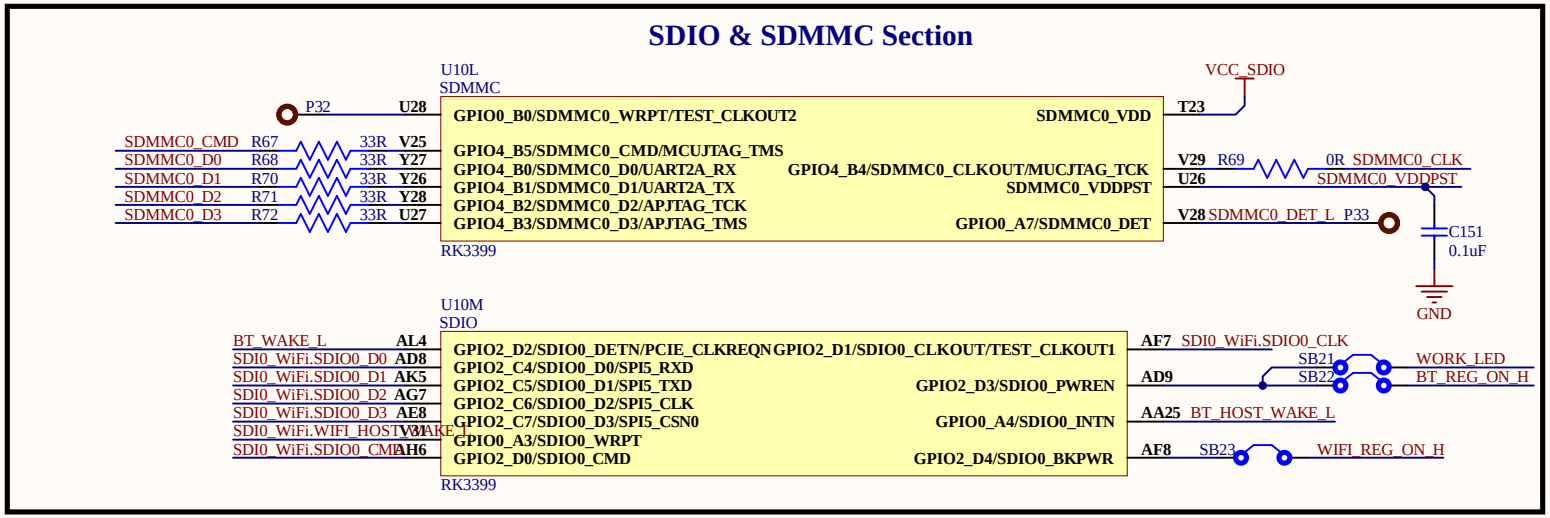
VCCA_0V9 — VCCA_0V9

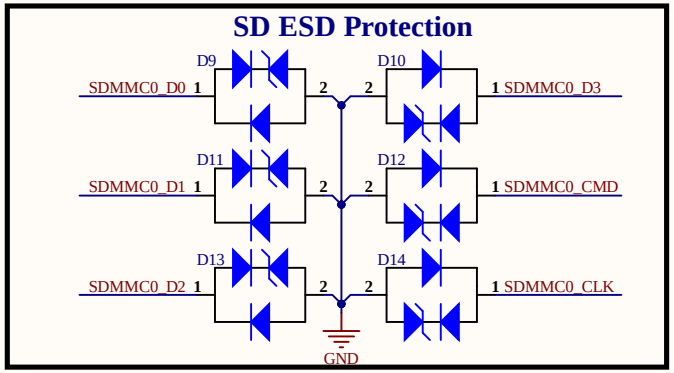
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	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 17 of 27	
	File: C:\Users\..\PCIe_Sheet.SchDoc	Drawn By: Huzaifa Ali	

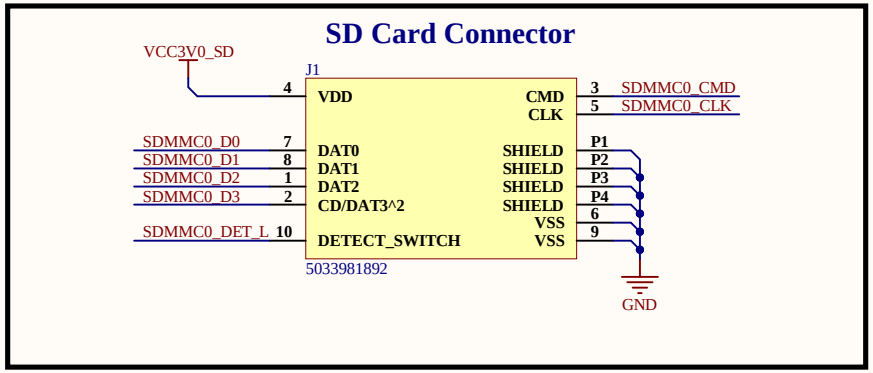




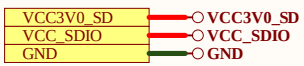




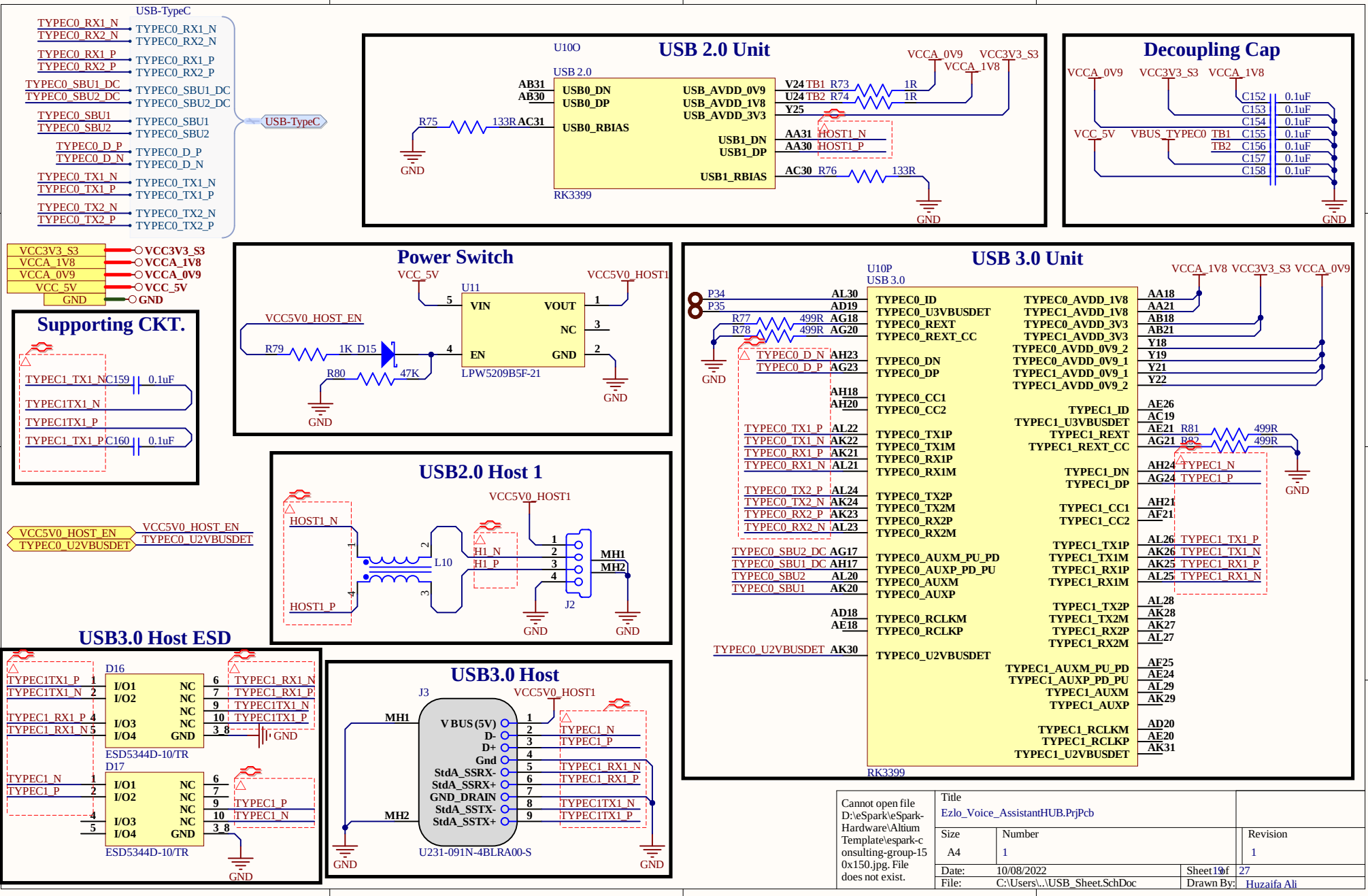


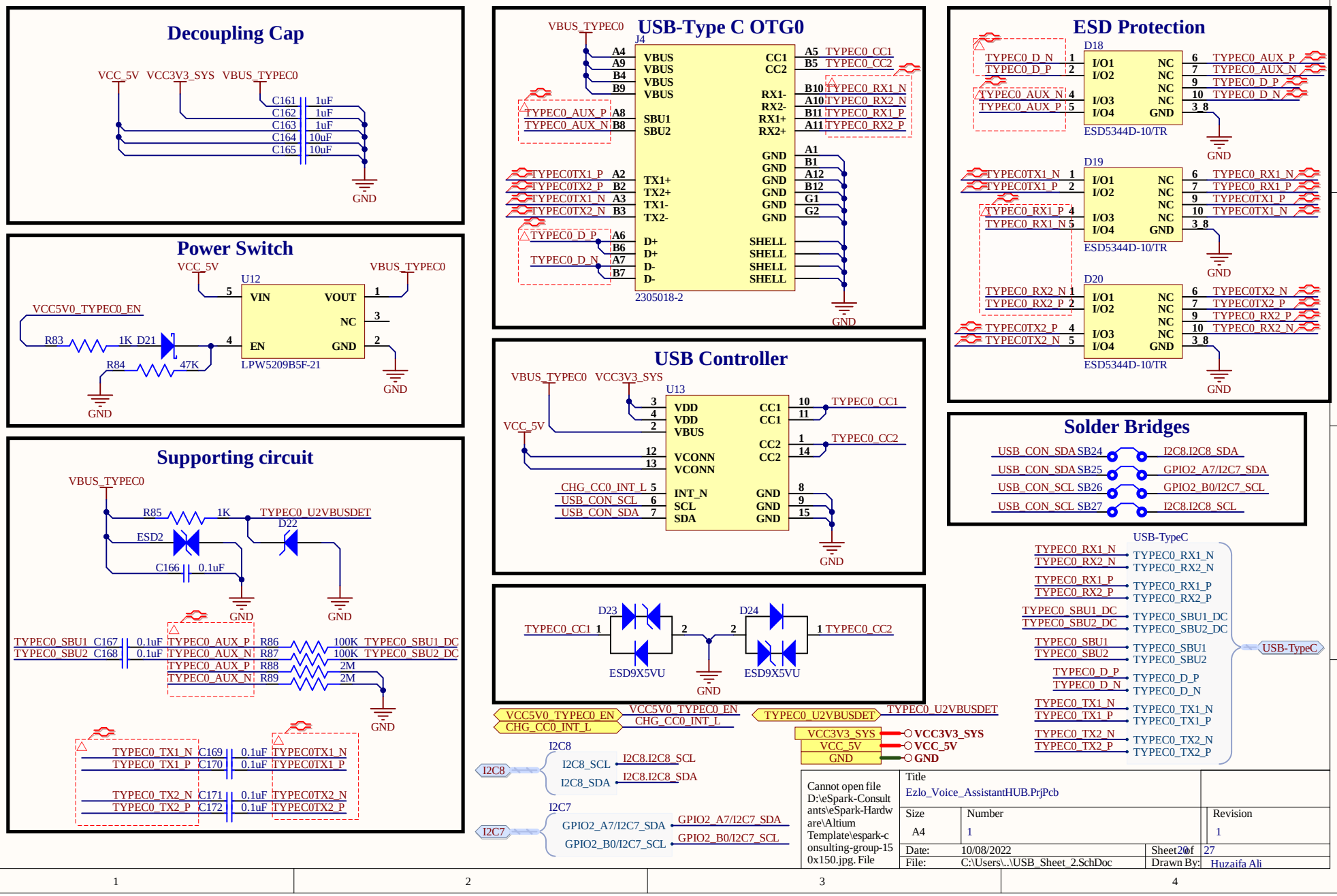


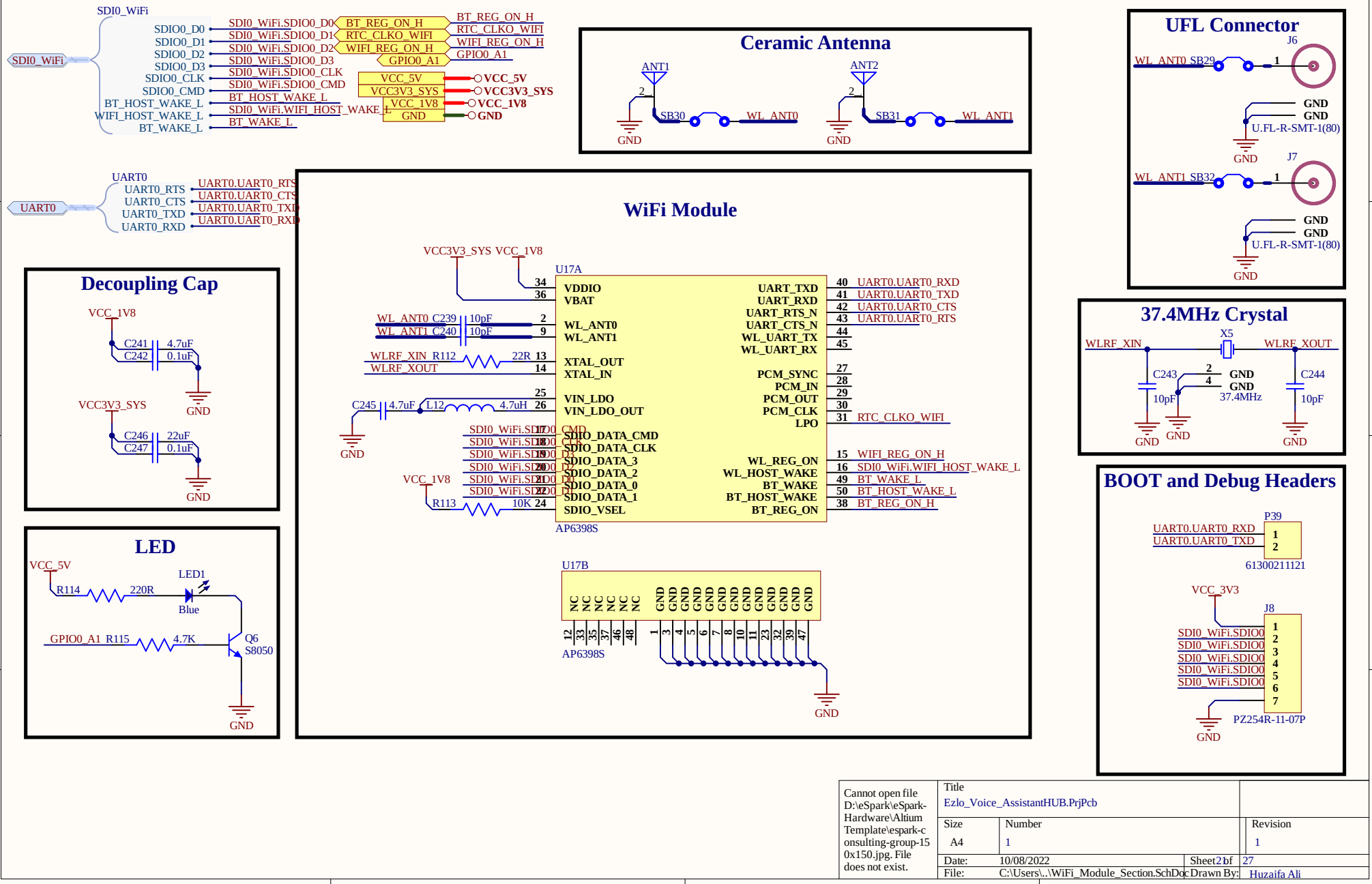


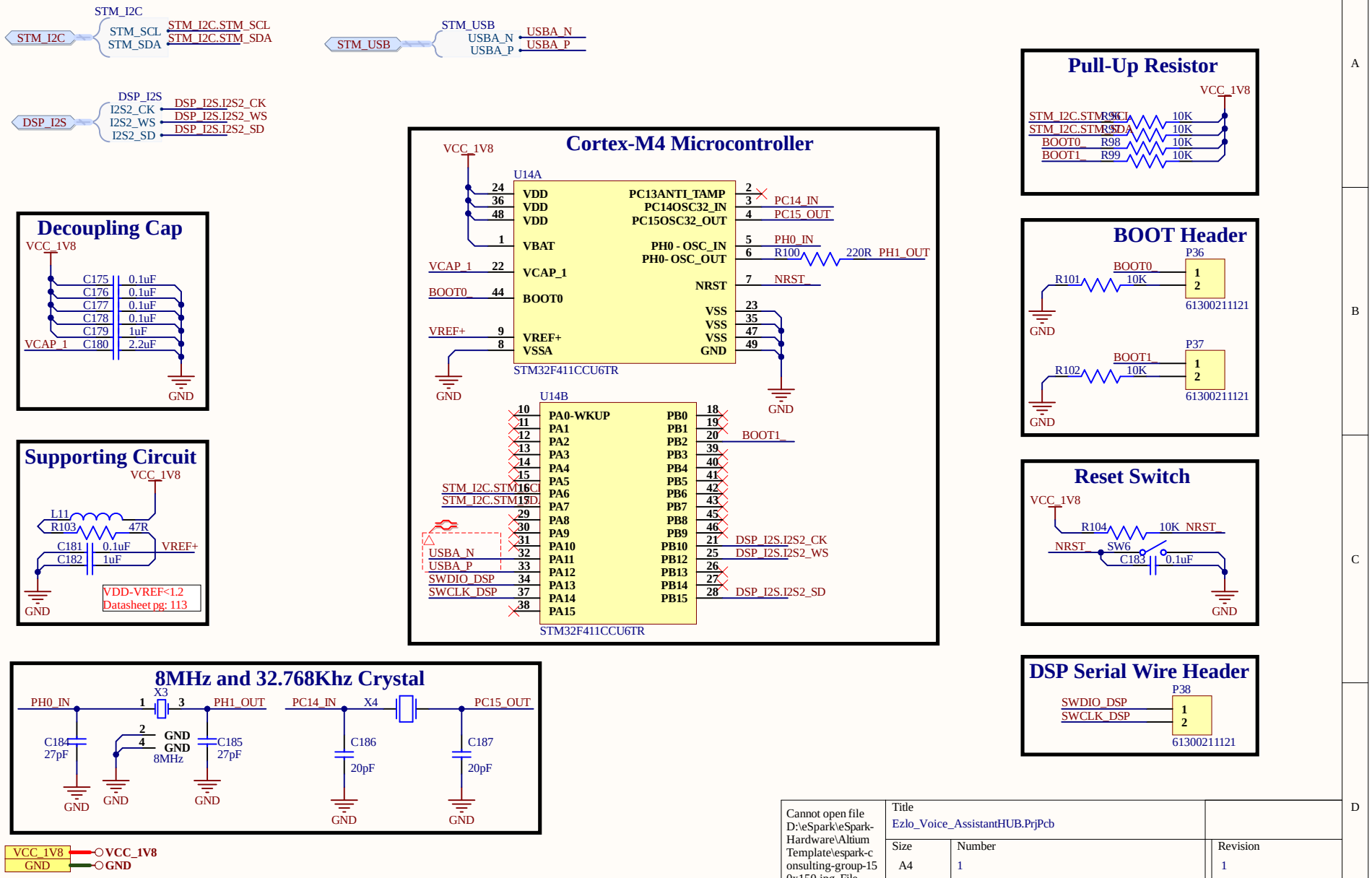


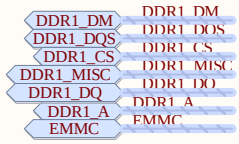
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_AssistantHUB.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 18 of 27	
	File: C:\Users\..\SDIO_SDMMC_Sheet.SchDoc	Drawn By: Huzaifa Ali	


USB-TypeC
TYPEC0_RX1_N
TYPEC0_RX2_N
TYPEC0_RX1_P
TYPEC0_RX2_P
TYPEC0_SBU1_DC
TYPEC0_SBU2_DC
TYPEC0_SBU1
TYPEC0_SBU2
TYPEC0_D_P
TYPEC0_D_N
TYPEC0_TX1_N
TYPEC0_TX1_P
TYPEC0_TX2_N
TYPEC0_TX2_P
USB-TypeC
USB 2.0 Unit
U10O
USB 2.0
AB31 USB0_DN
AB30 USB0_DP
AC31 USB0_RBIAS
USB_AVDD_0V9
USB_AVDD_1V8
USB_AVDD_3V3
USB1_DN
USB1_DP
USB1_RBIAS
V24 TB1 R73 1R
U24 TB2 R74 1R
Y25
AA31 HOST1_N
AA30 HOST1_P
AC30 R76 133R
R75 133R
RK3399
VCCA_0V9
VCC3V3_S3
VCCA_1V8
GND
Decoupling Cap
VCC_5V
VBUS_TYPEC0
C152 0.1uF
C153 0.1uF
C154 0.1uF
C155 0.1uF
C156 0.1uF
C157 0.1uF
C158 0.1uF
VCC3V3_S3
VCCA_1V8
VCCA_0V9
VCC_5V
GND
Power Switch
U11
VIN
VOUT
NC
EN
GND
LPW5209B5F-21
VCC5V0_HOST1
VCC5V0_HOST_EN
R79 1K D15
R80 47K
Supporting CKT.
TYPEC1_TX1_N C159 0.1uF
TYPEC1TX1_N
TYPEC1TX1_P
TYPEC1_TX1_P C160 0.1uF
USB 3.0 Unit
U10P
USB 3.0
P34
P35
R77 499R
R78 499R
TYPEC0_D_N AH23
TYPEC0_D_P AG23
TYPEC0_ID
TYPEC0_U3VBUSDET
TYPEC0_REXT
TYPEC0_REXT_CC
TYPEC0_DN
TYPEC0_DP
TYPEC0_CC1
TYPEC0_CC2
TYPEC0_TX1_P AL22
TYPEC0_TX1_N AK22
TYPEC0_RX1_P AK21
TYPEC0_RX1_N AL21
TYPEC0_TX1P
TYPEC0_TX1M
TYPEC0_RX1P
TYPEC0_RX1M
TYPEC0_TX2_P AL24
TYPEC0_TX2_N AK24
TYPEC0_RX2_P AK23
TYPEC0_RX2_N AL23
TYPEC0_TX2P
TYPEC0_TX2M
TYPEC0_RX2P
TYPEC0_RX2M
TYPEC0_SBU2_DC AG17
TYPEC0_SBU1_DC AH17
TYPEC0_SBU2 AL20
TYPEC0_SBU1 AK20
TYPEC0_AUXM_PU_PD
TYPEC0_AUXP_PD_PU
TYPEC0_AUXM
TYPEC0_AUXP
AD18 TYPEC0_RCLKM
AE18 TYPEC0_RCLKP
TYPEC0_U2VBUSDET AK30
TYPEC0_U2VBUSDET
TYPEC0_AVDD_1V8
TYPEC1_AVDD_1V8
TYPEC0_AVDD_3V3
TYPEC1_AVDD_3V3
TYPEC0_AVDD_0V9_2
TYPEC0_AVDD_0V9_1
TYPEC1_AVDD_0V9_1
TYPEC1_AVDD_0V9_2
TYPEC1_ID
TYPEC1_U3VBUSDET
TYPEC1_REXT
TYPEC1_REXT_CC
TYPEC1_DN
TYPEC1_DP
TYPEC1_CC1
TYPEC1_CC2
TYPEC1_TX1P
TYPEC1_TX1M
TYPEC1_RX1P
TYPEC1_RX1M
TYPEC1_TX2P
TYPEC1_TX2M
TYPEC1_RX2P
TYPEC1_RX2M
TYPEC1_AUXM_PU_PD
TYPEC1_AUXP_PD_PU
TYPEC1_AUXM
TYPEC1_AUXP
TYPEC1_RCLKM
TYPEC1_RCLKP
TYPEC1_U2VBUSDET
AA18
AA21
AB18
AB21
Y18
Y19
Y21
Y22
AE26
AC19
AE21 R81 499R
AG21 R82 499R
AH24 TYPEC1_N
AG24 TYPEC1_P
AH21
AF21
AL26 TYPEC1_TX1_P
AK26 TYPEC1_TX1_N
AK25 TYPEC1_RX1_P
AL25 TYPEC1_RX1_N
AL28
AK28
AK27
AL27
AF25
AE24
AL29
AK29
AD20
AE20
AK31
RK3399
VCCA_1V8
VCC3V3_S3
VCCA_0V9
GND
VCC5V0_HOST_EN
TYPEC0_U2VBUSDET
USB2.0 Host 1
HOST1_N
HOST1_P
L10
H1_N
H1_P
MH1
MH2
J2
VCC5V0_HOST1
USB3.0 Host ESD
D16
D17
I/O1
I/O2
I/O3
I/O4
NC
GND
ESD5344D-10/TR
TYPEC1TX1_P
TYPEC1TX1_N
TYPEC1_RX1_P
TYPEC1_RX1_N
TYPEC1_N
TYPEC1_P
TYPEC1TX1_N
TYPEC1TX1_P
USB3.0 Host
J3
VBUS (5V)
D-
D+
Gnd
StdA_SSRX-
StdA_SSRX+
GND_DRAIN
StdA_SSTX-
StdA_SSTX+
U231-091N-4BLRA00-S
MH1
MH2
TYPEC1_N
TYPEC1_P
TYPEC1_RX1_N
TYPEC1_RX1_P
TYPEC1TX1_N
TYPEC1TX1_P
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.
Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size
A4
Number
1
Revision
1
Date: 10/08/2022
Sheet 19 of 27
File: C:\Users\..\USB_Sheet.SchDoc
Drawn By: Huzaifa Ali


Decoupling Cap
VCC_5V VCC3V3_SYS VBUS_TYPEC0
C161 1uF
C162 1uF
C163 1uF
C164 10uF
C165 10uF
GND
Power Switch
VCC_5V
VBUS_TYPEC0
U12
VIN
VOUT
NC
EN
GND
LPW5209B5F-21
VCC5V0_TYPEC0_EN
R83 1K D21
R84 47K
Supporting circuit
VBUS_TYPEC0
R85 1K
TYPEC0_U2VBUSDET
D22
ESD2
C166 0.1uF
TYPEC0_SBU1_C167 0.1uF TYPEC0_AUX_P R86 100K TYPEC0_SBU1_DC
TYPEC0_SBU2_C168 0.1uF TYPEC0_AUX_N R87 100K TYPEC0_SBU2_DC
TYPEC0_AUX_P R88 2M
TYPEC0_AUX_N R89 2M
TYPEC0_TX1_N C169 0.1uF TYPEC0TX1_N
TYPEC0_TX1_P C170 0.1uF TYPEC0TX1_P
TYPEC0_TX2_N C171 0.1uF TYPEC0TX2_N
TYPEC0_TX2_P C172 0.1uF TYPEC0TX2_P
USB-Type C OTG0
VBUS_TYPEC0
J4
VBUS
CC1
CC2
RX1-
RX2-
RX1+
RX2+
SBU1
SBU2
TX1+
TX2+
TX1-
TX2-
D+
D-
SHELL
TYPEC0_AUX_P
TYPEC0_AUX_N
TYPEC0TX1_P
TYPEC0TX2_P
TYPEC0TX1_N
TYPEC0TX2_N
TYPEC0_D_P
TYPEC0_D_N
TYPEC0_CC1
TYPEC0_CC2
TYPEC0_RX1_N
TYPEC0_RX2_N
TYPEC0_RX1_P
TYPEC0_RX2_P
2305018-2
USB Controller
VBUS_TYPEC0 VCC3V3_SYS
U13
VDD
VBUS
VCONN
INT_N
SCL
SDA
CHG_CC0_INT_L
USB_CON_SCL
USB_CON_SDA
D23
D24
ESD9X5VU
ESD Protection
D18
D19
D20
I/O1
I/O2
I/O3
I/O4
NC
GND
ESD5344D-10/TR
Solder Bridges
USB_CON_SDA SB24
USB_CON_SDA SB25
USB_CON_SCL SB26
USB_CON_SCL SB27
I2C8.I2C8_SDA
GPIO2_A7/I2C7_SDA
GPIO2_B0/I2C7_SCL
I2C8.I2C8_SCL
USB-TypeC
VCC5V0_TYPEC0_EN
CHG_CC0_INT_L
TYPEC0_U2VBUSDET
VCC3V3_SYS
VCC_5V
GND
I2C8
I2C8_SCL
I2C8_SDA
I2C7
GPIO2_A7/I2C7_SDA
GPIO2_B0/I2C7_SCL
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File
Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size
A4
Number
1
Revision
1
Date: 10/08/2022
Sheet 20 of 27
File: C:\Users\..\USB_Sheet_2.SchDoc
Drawn By: Huzaifa Ali


SDI0_WiFi
SDIO0_D0
SDIO0_D1
SDIO0_D2
SDIO0_D3
SDIO0_CLK
SDIO0_CMD
BT_HOST_WAKE_L
WIFI_HOST_WAKE_L
BT_WAKE_L
SDI0_WiFi.SDIO0_D0
SDI0_WiFi.SDIO0_D1
SDI0_WiFi.SDIO0_D2
SDI0_WiFi.SDIO0_D3
SDI0_WiFi.SDIO0_CLK
SDI0_WiFi.SDIO0_CMD
BT_HOST_WAKE_L
SDI0_WiFi.WIFI_HOST_WAKE_L
BT_WAKE_L
BT_REG_ON_H
RTC_CLKO_WIFI
WIFI_REG_ON_H
GPIO0_A1
VCC_5V
VCC3V3_SYS
VCC_1V8
GND
BT_REG_ON_H
RTC_CLKO_WIFI
WIFI_REG_ON_H
GPIO0_A1
VCC_5V
VCC3V3_SYS
VCC_1V8
GND
Ceramic Antenna
ANT1
ANT2
SB30
SB31
WL_ANT0
WL_ANT1
GND
UFL Connector
J6
J7
WL_ANT0_SB29
WL_ANT1_SB32
GND
U.FL-R-SMT-1(80)
UART0
UART0_RTS
UART0_CTS
UART0_TXD
UART0_RXD
UART0.UART0_RTS
UART0.UART0_CTS
UART0.UART0_TXD
UART0.UART0_RXD
WiFi Module
U17A
VCC3V3_SYS
VCC_1V8
VDDIO
VBAT
WL_ANT0
WL_ANT1
XTAL_OUT
XTAL_IN
VIN_LDO
VIN_LDO_OUT
SDIO_DATA_CMD
SDIO_DATA_CLK
SDIO_DATA_3
SDIO_DATA_2
SDIO_DATA_0
SDIO_DATA_1
SDIO_VSEL
UART_TXD
UART_RXD
UART_RTS_N
UART_CTS_N
WL_UART_TX
WL_UART_RX
PCM_SYNC
PCM_IN
PCM_OUT
PCM_CLK
LPO
WL_REG_ON
WL_HOST_WAKE
BT_WAKE
BT_HOST_WAKE
BT_REG_ON
WL_ANT0 C239 10pF
WL_ANT1 C240 10pF
WLRF_XIN R112 22R
WLRF_XOUT
C245 4.7uF L12 4.7uH
R113 10K
AP6398S
U17B
AP6398S
37.4MHz Crystal
X5
WLRF_XIN
WLRF_XOUT
C243 10pF
C244 10pF
37.4MHz
Decoupling Cap
C241 4.7uF
C242 0.1uF
C246 22uF
C247 0.1uF
LED
VCC_5V
R114 220R
LED1
Blue
GPIO0_A1 R115 4.7K
Q6
S8050
BOOT and Debug Headers
P39
UART0.UART0_RXD
UART0.UART0_TXD
61300211121
VCC_3V3
J8
SDI0_WiFi.SDIO0
PZ254R-11-07P
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.
Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size
A4
Number
1
Revision
1
Date:
10/08/2022
Sheet 2 of 27
File:
C:\Users\..\WiFi_Module_Section.SchDoc
Drawn By:
Huzaifa Ali


STM_I2C
STM_SCL
STM_SDA
STM_I2C.STM_SCL
STM_I2C.STM_SDA
STM_USB
USBA_N
USBA_P
DSP_I2S
I2S2_CK
I2S2_WS
I2S2_SD
DSP_I2S.I2S2_CK
DSP_I2S.I2S2_WS
DSP_I2S.I2S2_SD
Pull-Up Resistor
VCC_1V8
R96 10K
R97 10K
R98 10K
R99 10K
BOOT0_
BOOT1_
Decoupling Cap
C175 0.1uF
C176 0.1uF
C177 0.1uF
C178 0.1uF
C179 1uF
C180 2.2uF
VCAP_1
GND
Cortex-M4 Microcontroller
U14A
VDD
VBAT
VCAP_1
BOOT0
VREF+
VSSA
PC13ANTI_TAMP
PC14OSC32_IN
PC15OSC32_OUT
PH0 - OSC_IN
PH0- OSC_OUT
NRST
VSS
PC14_IN
PC15_OUT
PH0_IN
R100 220R PH1_OUT
NRST_
STM32F411CCU6TR
U14B
PA0-WKUP
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
BOOT1_
DSP_I2S.I2S2_CK
DSP_I2S.I2S2_WS
DSP_I2S.I2S2_SD
USBA_N
USBA_P
SWDIO_DSP
SWCLK_DSP
BOOT Header
P36
P37
R101 10K
R102 10K
61300211121
Supporting Circuit
L11
R103 47R
C181 0.1uF
C182 1uF
VREF+
VDD-VREF<1.2
Datasheet pg: 113
Reset Switch
R104 10K NRST
SW6
C183 0.1uF
8MHz and 32.768Khz Crystal
X3
X4
PH0_IN
PH1_OUT
PC14_IN
PC15_OUT
C184 27pF
C185 27pF
C186 20pF
C187 20pF
8MHz
DSP Serial Wire Header
P38
SWDIO_DSP
SWCLK_DSP
VCC_1V8
GND
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Title
Ezlo_Voice_AssistantHUB.PrjPcb
Size A4
Number 1
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Date: 10/08/2022
Sheet 22 of 27
File: C:\Users\..\DSP_Section.SchDoc
Drawn By: Huzaifa Ali



U_RAM_Sheet
RAM_Sheet.SchDoc

DDR1_DM
DDR1_DQS
DDR1_CS
DDR1_MISC
GND
VCC_DDR
VCC_1V8

U_RAM_Sheet_2
RAM_Sheet_2.SchDoc

DDR1_DQ
DDR1_A
GND

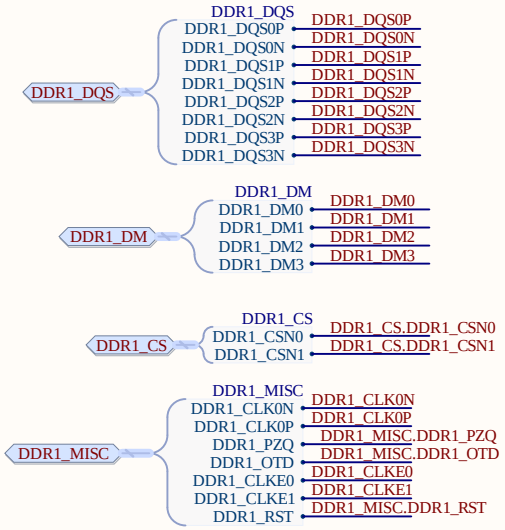
U_Flash_Sheet
Flash_Sheet.SchDoc

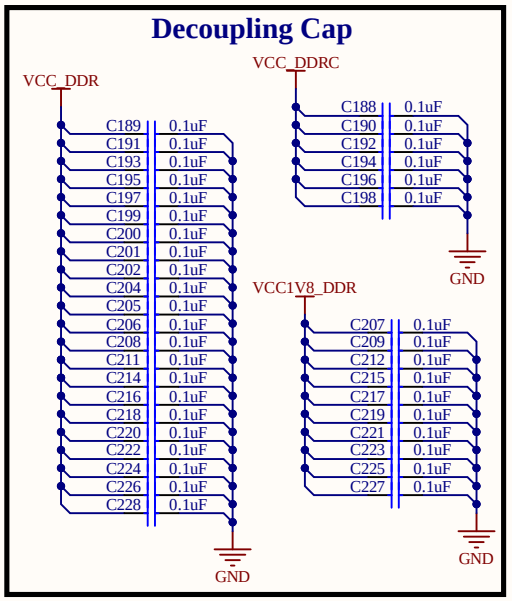
VCC_1V8
GND
EMMC
VCC3V3_S3

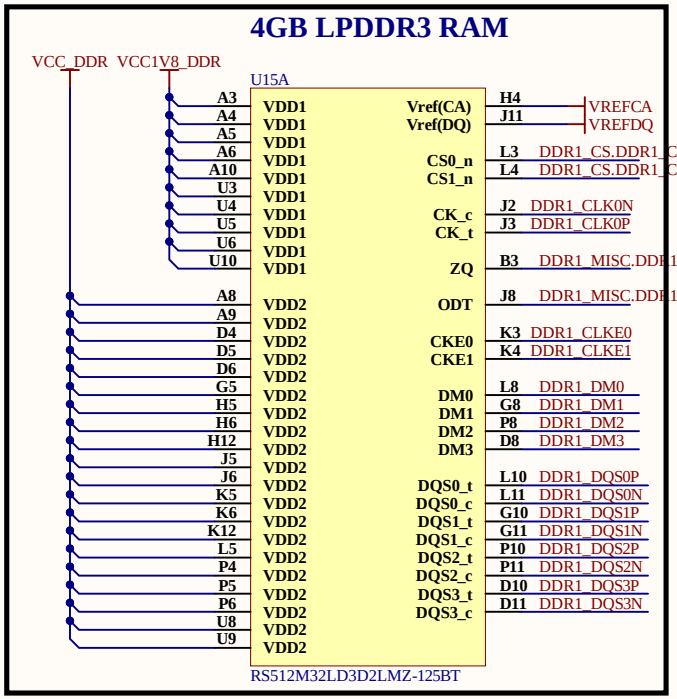
VCC_DDR
VCC_1V8
VCC3V3_S3
GND

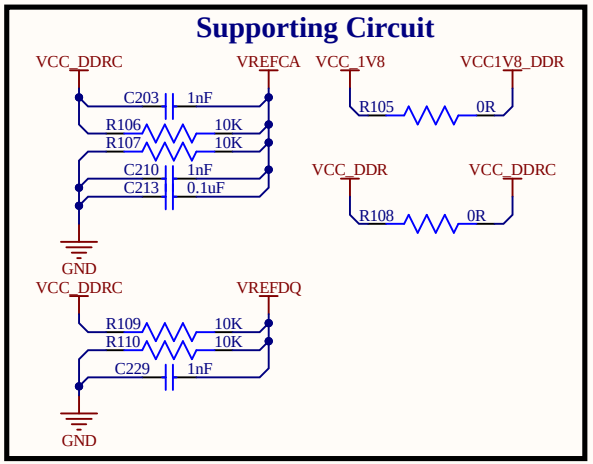
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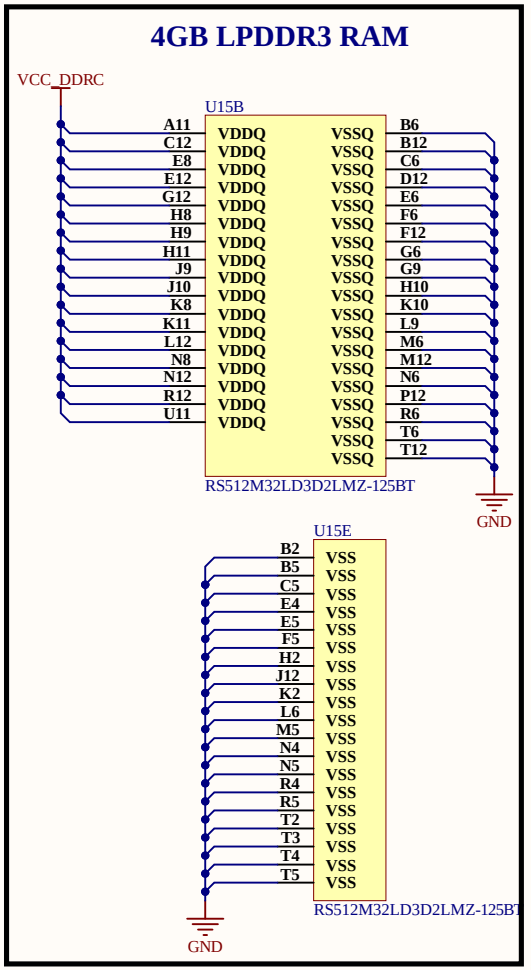
Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	10/08/2022	Sheet 22 of	27
File:	C:\Users\..\Memory_Section.SchDoc	Drawn By:	Huzaifa Ali

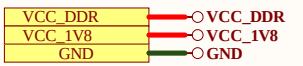




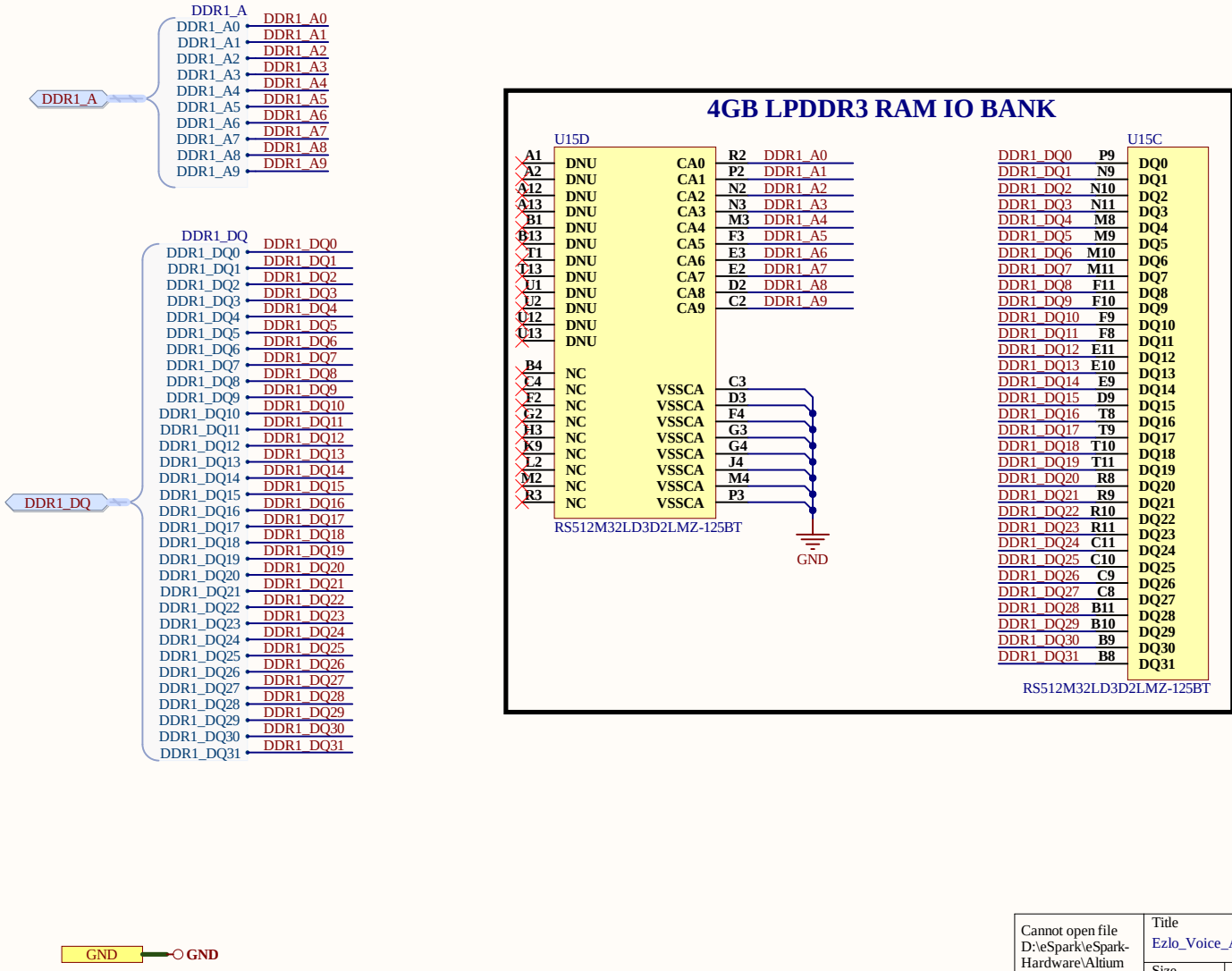




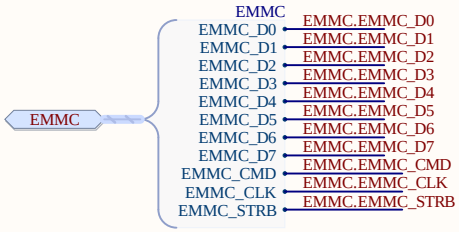


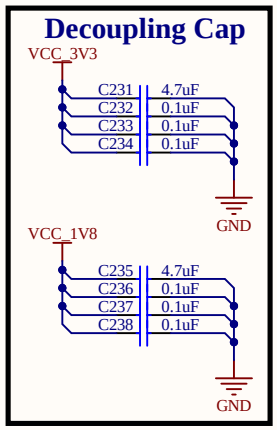


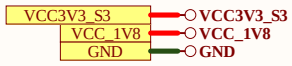
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	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 24 of 27	
	File: C:\Users\..\RAM_Sheet.SchDoc	Drawn By: Huzaifa Ali	

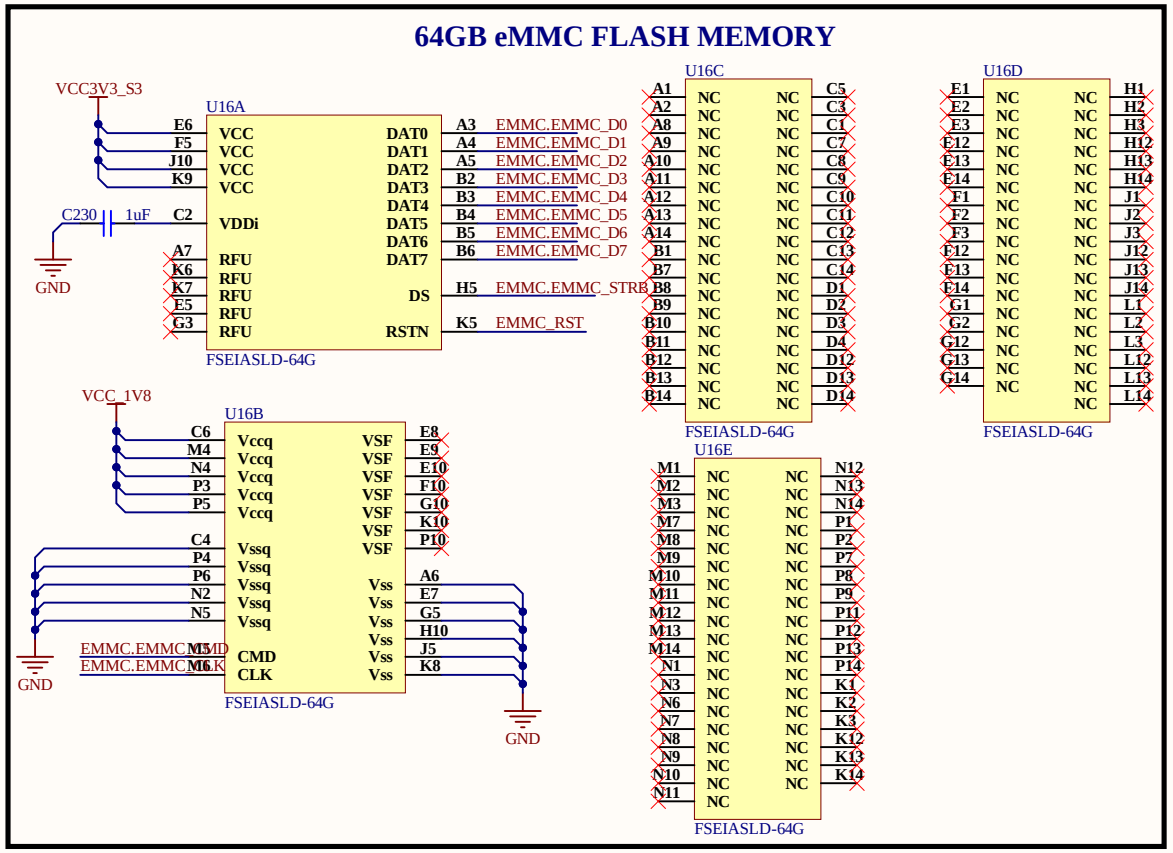


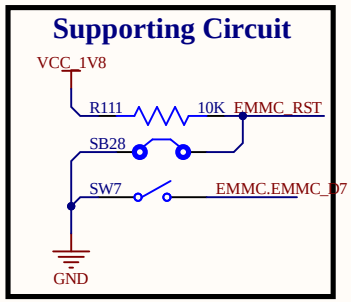
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	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 25 of 27	
	File: C:\Users\..\RAM_Sheet_2.SchDoc	Drawn By: Huzaifa Ali	









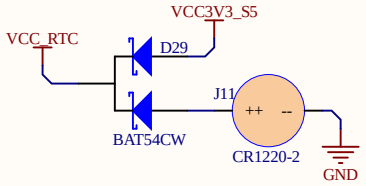


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	Size: A4	Number: 1	Revision: 1
	Date: 10/08/2022	Sheet 26 of 27	
	File: C:\Users\..\Flash_Sheet.SchDoc	Drawn By: Huzaifa Ali	

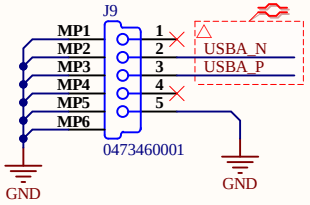
UART2DBG
UART2DBG_RX
UART2DBG_TX
UART2DBG.UART2DBG_RX
UART2DBG.UART2DBG_TX

STM_USB
USBA_N
USBA_P
USBA_N
USBA_P

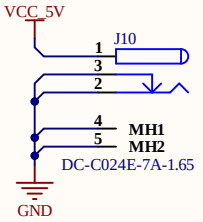
RTC Battery



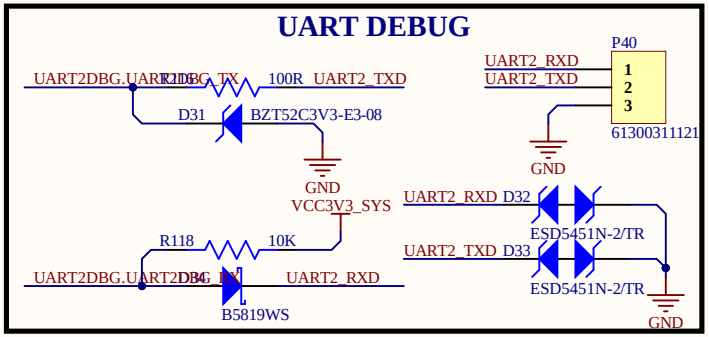
STM PROGRAMMING USB



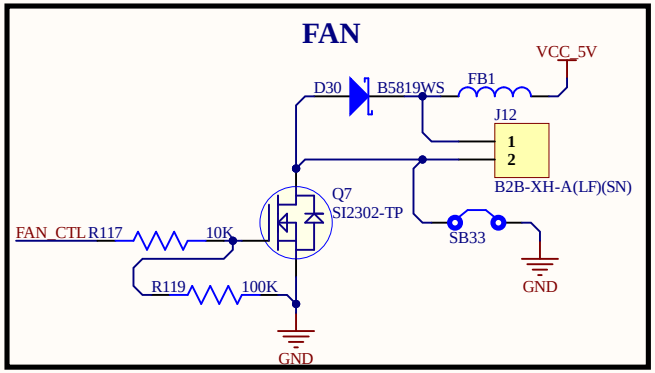
INPUT JACK



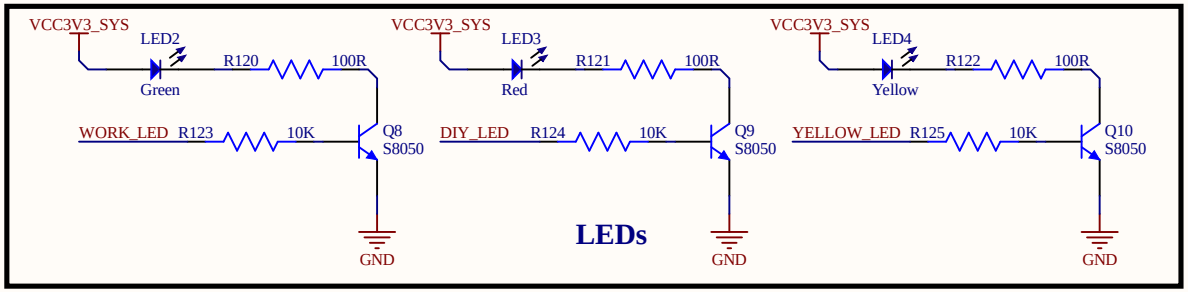
UART DEBUG



FAN



LEDs



YELLOW_LED
DIY_LED
WORK_LED
YELLOW_LED
DIY_LED
WORK_LED

FAN_CTL
FAN_CTL

VCC3V3_SYS
VCC3V3_S5
VCC_RTC
VCC_5V
GND
VCC3V3_SYS
VCC3V3_S5
VCC_RTC
VCC_5V
GND

Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	Ezlo_Voice_AssistantHUB.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	10/08/2022	Sheet 27 of	27
File:	C:\Users\..\Connector_Section.SchDoc	Drawn By:	Huzaifa Ali